

Degree-Weighted Copy Certificates for Sparse Modularity Contraction

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ABSTRACT

Safe contraction retains a globally optimal clustering while reducing its search space. Modularity makes local contraction difficult because its degree-based null model is dense. We copy a candidate block to an existing community through an anchor chosen proportional to degree, cancelling the exterior null interaction exactly. Sparse normalized adjacency profiles then give a collective spectral certificate: a nonnegative restriction preserves an optimal fused representative, and a positive restriction forces fusion in every optimum. Exact rational checks certify acceptance, with compatible overlap and a quantified loss bound. The same copy argument preserves a plain metric semidefinite relaxation with all triangle constraints. On six public graphs at three resolutions, a degree stage after recursive criteria adds removals in 44 of 54 fixed-seed cases and 14 of 18 setting medians; sequential median removal is 1.79–49.54% of vertices. Preparation does not amortize in the measured Louvain workloads: all 54 graph-resolution-workload median speedups against unreduced computation stay below one. In a separate, exactly audited 43-case metric-SDP comparison, SD and the stronger SW control both verify 36 targets; SD is faster in 15 and slower in 21 jointly verified pairs. The conditional median complete-cost ratio T_{SD}/T_{SW} is 1.0075. Controlled discrete audits expose additional exact proofs, while every public optimum remains unresolved under the capped policy.

KEYWORDS

graph clustering, modularity, safe contraction, sparse matrices, partial optimality

1 INTRODUCTION

Graph clustering combines a local representation with a global objective. Modularity compares observed within-community adjacency with a degree-based null model [3, 4]. Its objective matrix is dense: even two nonadjacent vertices have a nonzero interaction. Contracting vertices before optimization can reduce the search space, but a useful contraction should retain an optimal partition rather than merely reproduce a preliminary heuristic partition.

Partial optimality supplies this guarantee through transformations that improve an objective for every feasible partition. Existing pair rules, collective almost-clique conditions, and correlation-clustering preprocessing already provide safe fusions [6–8, 10]. Their collective criteria also offer a necessary standard of comparison: certifying a block that defeats one pair rule does not establish an advantage over recursive preprocessing. Our question is whether the specific structure of modularity permits a collective certificate whose exterior calculation uses sparse adjacency profiles.

The difficulty is the exterior assignment. When a block is copied to one community, every vertex in that block may change its relationship with vertices outside the block. A uniform choice of the destination anchor compares rows of the modularity matrix. Degree heterogeneity leaves a dense null-model difference in those rows. We instead choose an anchor in proportion to its degree. For two block vertices u, v and an exterior vertex w , the resulting comparison is

$$d_v B_{uw} - d_u B_{vw} = d_v A_{uw} - d_u A_{vw}. \quad (1)$$

The null-model terms cancel. The exterior cost of copying the block can thus be bounded with differences of normalized sparse adjacency rows. Uniform-anchor contrasts also admit sparse evaluation by summing the rank-one term on the complement of the observed neighbor union; degree weighting eliminates that term and yields the normalized-profile bound used here.

We turn this cancellation into a restricted spectral test. Internal connectivity must compensate for exterior profile variation and the resolution-dependent cost of bringing the block together. A nonnegative restricted matrix preserves at least one global optimum; a positive definite restriction forces the block together in every optimum. The proof also explains why weak certificates from the same graph can overlap safely: at a global optimum, every anchor of positive probability preserves optimality. This property is stronger than existential weak persistency alone.

The contribution is a degree-weighted collective certificate and a sparse, exactly verified implementation of it. Quotient preservation, graph contraction, and general improving mappings are established components. We examine the certificate’s incremental coverage against the specified pair, collective and recursive criteria, and distinguish sparse exterior assembly from the cost of exact matrix verification. Public coverage is complementary to the specified recursive criteria, including on their actual quotient, but the measured Louvain workloads retain complete-cost overhead. We also prove that the copy certificate preserves an optimal fused representative of a metric semidefinite relaxation. A separate fixed comparison measures complete bound-computation cost against optimized pair/uniform and stronger full-pair degree-anchor controls at an exact original-normalized width target, finding no robust advantage over the full-pair control. Structural coverage and optimization utility therefore have distinct endpoints. A connected Paley-family construction separates joint degree-specific collective fusion from the named original predicates; its surviving isolation-and-uniform composition limits that distinction.

2 BACKGROUND AND CLOSEST WORK

2.1 Modularity and degree-preserving quotients

Let $A = A^\top \geq 0$ be an adjacency matrix, including any diagonal entries. Write $d = A\mathbf{1}$ and $S = \mathbf{1}^\top d > 0$. For resolution $\gamma \geq 0$, define

$$B = A - \gamma dd^\top / S, \quad Q_\gamma(P) = S^{-1} \sum_{u,v} B_{uv} \mathbf{1}[P(u) = P(v)]. \quad (2)$$

We allow every set partition, with no fixed cluster count or balance constraint. The diagonal in Eq. (2) is constant, so maximizing Q_γ is equivalent to maximizing $F(P) = \sum_{u < v} B_{uv} \mathbf{1}[P(u) = P(v)]$.

For a partition of vertices into contracted groups, let R be its indicator matrix. The established quotient is $A' = R^\top A R$ [5]. It retains $d' = R^\top d$ and $S' = S$, and every quotient partition has the same modularity as its lift. In particular, A'_{aa} contains twice the off-diagonal adjacency mass inside group a . These diagonal entries are necessary when degrees are recomputed. Our certificate determines which groups can be contracted safely; the quotient identity itself is not a new result. Arenas et al. also give safe local fusion rules for hairs and triangular fringes at ordinary modularity ($\gamma = 1$), subject to explicit loop and degree-volume conditions. Including a triangular fringe's common neighbor requires a further condition; quotient preservation alone does not justify that inclusion. Peripheral-clique filtering has also been studied for modularity ILPs [2]. We require a proved block condition rather than inferring optimum preservation from finite validation of a filtering rule.

2.2 Improving maps and collective preprocessing

Correlation clustering has an equivalent disagreement objective after the sign and magnitude of each B_{uv} are interpreted as pair preferences. Improving-map methods provide pair and subgraph conditions for partial optimality [6, 7]. General relaxed improving mappings are also established [30]. The exact similar-neighborhood rule already covers the two-vertex specialization of our copy bound [8]; Appendix B gives the implication. Our distinct criterion concerns collective blocks and sparse profile computation, rather than a stronger pair rule. The strict positive-closure edge criterion tests whether a positive pair dominates its competing positive incident edges. The almost-clique criterion compares the positive internal minimum cut with negative internal mass and positive exterior mass [8]. Both consider collective structure beyond graph twins.

Modern exact cluster-editing solvers combine several such reductions recursively [10]. A C-subinstance condition goes further: it optimizes the objective incident to a supplied block and checks whether isolating that block attains the subinstance optimum. Its optimization cost and its tie handling differ from a local matrix certificate. We compare the actual isolation score, rather than inferring its condition from one returned partition. Full exact solving is a separate comparison from selected preprocessing rules.

Uniform-anchor copying yields a useful reference. If $D_{uv} = \sum_{w \notin K} |B_{uw} - B_{vw}|$, positivity of the signed Laplacian with weights $B_{uv} - D_{uv}/|K|$ suffices for fusion. Degree weighting changes the exterior contrast to Eq. (1); it does not uniformly dominate the

uniform test. Normalized modularity spectral analysis relates resolution to the Fiedler space [14]. Convexified modularity recovery already uses the internal signed Laplacian $L(B_K)$ in its planted-partition dual construction [25]. Our supplied-block condition instead bounds observed arbitrary exterior assignments. Its zero-exterior limit recovers the established normalized-Laplacian threshold. For isolated components at $\gamma = 1$, this eigenvalue test is a sufficient relaxation of Louf et al.'s exact normalized-cut non-splitting threshold [1]; Appendix D gives the comparison.

Spectral proximity can certify global RatioCut or NCut optimality through an exact semidefinite relaxation [24]. Our fixed-instance guarantee concerns a local contraction for unrestricted discrete modularity.

3 A DEGREE-WEIGHTED COLLECTIVE CERTIFICATE

3.1 Copying a block to an anchor's community

Fix a block K of $k \geq 2$ vertices with $d_u > 0$ for every $u \in K$. Let A_K denote its principal adjacency submatrix and d_K its original full-degree vector restricted to K . Let $V = \sum_{u \in K} d_u$. For an anchor $r \in K$, the map $T_{K,r}$ places all of K into r 's current community and leaves every exterior vertex fixed. Choose r with probability d_r/V .

This distribution is determined by the cancellation requirement. For any positive anchor probabilities π , the exterior null contrast is proportional to $d_w(\pi_v d_u - \pi_u d_v)$. At nonzero resolution, if an exterior vertex has positive degree, cancellation of every such contrast forces $\pi_u = d_u/V$. This characterizes exact cancellation, rather than the best distribution for certification; a noncancelling distribution may still give a useful bound.

If the current partition separates $u, v \in K$, their pair becomes internal after every anchor copy. For the exterior, write $x_u^w = \mathbf{1}[P(u) = P(w)]$. Direct expansion gives

$$-\mathbb{E}_r \Delta F_{\text{ext}} = V^{-1} \sum_{u < v \in K} \sum_{w \notin K} (d_v B_{uw} - d_u B_{vw})(x_u^w - x_v^w), \quad (3)$$

$$P_{uv}^{\text{ext}} = V^{-1} \sum_{w \notin K} |d_v A_{uw} - d_u A_{vw}|. \quad (4)$$

When u, v share a community, their exterior summand vanishes. Otherwise its absolute value is bounded by Eq. (4). Therefore

$$\mathbb{E}_r \{F(T_{K,r}P) - F(P)\} \geq \sum_{\substack{u < v \in K \\ P(u) \neq P(v)}} (B_{uv} - P_{uv}^{\text{ext}}). \quad (5)$$

This expression retains negative internal pair weights.

For symmetric pair weights w , let $L(w) = \sum_{u < v} w_{uv}(e_u - e_v)(e_u - e_v)^\top$ be their signed Laplacian. If $K_1, \dots, K_t$ are the nonempty groups induced by P inside K , the right side of Eq. (5) equals $\frac{1}{2} \sum_j \mathbf{1}_{K_j}^\top L(B - P^{\text{ext}}) \mathbf{1}_{K_j}$. Positive semidefiniteness controls every such partition simultaneously. Selecting a maximum-gain anchor for each partition defines an improving map whenever this weighted mean is nonnegative. The resulting persistency conclusion therefore uses the established improving-map framework [7]; the matrix bound below supplies a computable modularity-specific sufficient condition.

3.2 A sparse normalized-row center

The exact pair penalty requires all pairs of exterior profiles. To reduce assembly cost, choose a fixed nonnegative exterior vector c , and define

$$\delta_u = \sum_{w \notin K} |A_{uw}/d_u - c_w|, \quad \bar{\delta} = V^{-1} \sum_{u \in K} d_u \delta_u. \quad (6)$$

Eq. (6) and triangle inequality give $P_{uv}^{\text{ext}} \leq (d_u d_v / V)(\delta_u + \delta_v)$. Let $W = \text{diag}(d_K)$ and let

$$M = W^{-1/2} L(A_K) W^{-1/2} - \text{diag}(\delta) - \bar{\delta} I. \quad (7)$$

Write λ for its least Rayleigh quotient over $\{z : z^\top \sqrt{d_K} = 0, \|z\| = 1\}$.

THEOREM 1 (COLLECTIVE FUSION). *Under the above assumptions, $\lambda \geq \gamma V / S$ implies that some global maximizer of modularity fuses all vertices of K . If $\lambda > \gamma V / S$, every global maximizer fuses K .*

PROOF. On the complementary hyperplane $d_K^\top x = 0$, the relaxed penalty has quadratic form $\sum_u d_u (\delta_u + \bar{\delta}) x_u^2$, while the internal null-model Laplacian contributes $-(\gamma V / S) \sum_u d_u x_u^2$. With $z = W^{1/2} x$, the contrast form is thus bounded below by $(\lambda - \gamma V / S) \|z\|^2$. The full contrast Laplacian kills $\mathbf{1}$, so translating an indicator along $\mathbf{1}$ puts it in $d_K^\perp$ without changing its energy. Equation (5) is nonnegative for every partition under the weak condition. At an optimum, at least one anchor copy consequently preserves the objective and fuses K . Under the strict condition, any nontrivial split of K has a positive indicator energy, contradicting optimality. Appendix A gives the expansion. $\square$

Internal connectivity, exterior profile dispersion, and resolution appear separately in Eq. (7). The internal matrix uses the original full degrees, rather than degrees inside the induced block. Two vertices with the same normalized exterior profile pay zero pair penalty, even when their absolute exterior weights differ. Conversely, a heterogeneous normalized profile can invalidate a contraction despite dense internal adjacency.

3.3 Resolution, overlap, and quantified loss

For a fixed adjacency-derived center, λ is independent of resolution. Theorem 1 therefore gives a threshold $\gamma_{\max} = S\lambda/V$: weak fusion holds at and below that threshold, and strict fusion holds below it. This is a family of optimal fused representatives, one for each resolution; it need not be one common partition. Each proposed rational endpoint is rechecked with Eq. (11); computed eigenvalues and numerical threshold estimates remain diagnostics.

LEMMA 1 (EVERY OPTIMAL ANCHOR). *For a block satisfying a nonnegative expected-copy bound at every partition, every positive-probability anchor maps every global optimum to a global optimum.*

PROOF. Every feasible move from an optimum has gain at most zero. The weighted mean of these gains is nonnegative. Every term with positive weight is therefore zero. $\square$

COROLLARY 1 (COMPATIBLE OVERLAP). *Blocks verified by Theorem 1 on the same graph at the same resolution are jointly weakly persistent. Each connected union of intersecting blocks can be fused simultaneously in a global optimum.*

Start with a certified block, then traverse intersecting blocks and choose each new anchor inside the already fused union. Lemma 1 preserves the optimum and the previous union. All block vertices must have positive degree; an isolated shared vertex has zero anchor probability and cannot justify this step. Generic weakly persistent constraints need not be jointly compatible.

The same calculation quantifies a failed test. Put $\mu = \lambda - \gamma V / S$, and let $v_j = \sum_{u \in K_j} d_u$. Then

$$\mathbb{E}_r \Delta F \geq \frac{\mu}{2} \left(V - \sum_j v_j^2 / V \right). \quad (8)$$

The loss between the unrestricted modularity optimum and the best partition that fuses K is at most

$$\frac{\max(0, -\mu)}{S} \left(V - \frac{\sum_{u \in K} d_u^2}{V} \right). \quad (9)$$

Loss bounds for disjoint original blocks add under successive anchor copies. The threshold and the loss bound are attained on complete graphs (Appendix A.5); they cannot be uniformly tightened using only the stated margin and degree dispersion. We use exact certificates for production contraction; a diagnostic negative margin alone does not authorize a merge. Transporting the same center through a coarsening that respects the block boundary can only strengthen this condition. Recomputing a median can instead weaken it, even after a safe coarsening; Appendix A.4 proves the transport statement and gives an exact counterexample.

4 SPARSE ASSEMBLY AND EXACT VERIFICATION

4.1 Computing exterior dispersion

Let $N(u) = \{v : A_{uv} > 0\}$, $m_{\text{ext}} = \sum_{u \in K} |N(u) \setminus K|$, and $\ell_K = \sum_{u \in K} \text{nnz}(A[u, :])$. For $e_u = \sum_{w \notin K} A_{uw}$ and $C = \sum_{w \notin K} c_w$, non-negativity gives

$$\delta_u = e_u / d_u + C - 2 \sum_{w \in N(u) \setminus K} \min(A_{uw} / d_u, c_w). \quad (10)$$

This identity avoids scanning the center support separately for every vertex. Our main checker chooses an unweighted coordinatewise median of the k normalized rows, counting missing entries as zero. A positive median requires nonzero support in at least half the rows. Extracting internal and exterior entries scans the incident CSR rows in $O(\ell_K)$ time. Sorting the exterior column lists then costs $O(m_{\text{ext}} \log k)$. Given the center, Eq. (10) takes $O(k + m_{\text{ext}} + |\text{supp } c|)$. Zero and fixed-anchor centers are ablations. A degree-weighted median minimizes weighted total dispersion, but that scalar objective does not by itself maximize the restricted spectral margin. Appendix G bounds the center's spectral overestimate and gives the weighted-center variant; production remains unweighted.

Sparse assembly removes a k -by-exterior dense table; it does not remove the internal matrix verification cost. Floating restricted eigensolving costs $O(k^3)$ in our dense implementation. Rational factorization has cubic arithmetic-operation count with additional integer bit cost. The configured production checker accepts blocks

of at most 64 vertices. Full-graph degree preparation and rational caching are charged separately before block checks. The current floating diagnostic uses generic CSR internal-column slicing, which also allocates a graph-width index workspace. The incident-only cost bound concerns exterior profile construction; measurements retain this additional implementation cost.

4.2 Separating numerical screening from certificates

Floating eigenvalues screen candidates. A screened failure is an abstention; floating positivity never certifies a merge. Every accepted block is checked with exact rational arithmetic. For a positive-degree pivot p , use columns $e_i - (d_i/d_p)e_p$ to form a rational basis U for $d_K^\perp$. Verify

$$U^\top [L(A_K) - \text{diag}\{d_u(\delta_u + \bar{\delta} + \gamma V/S)\}] U \geq 0. \quad (11)$$

Exact symmetric elimination distinguishes a zero pivot with a zero remaining row from a zero pivot with a nonzero row; the latter violates positive semidefiniteness. All positive pivots certify the strict case.

Inputs are symmetric nonnegative sparse matrices with positive total volume. The CSR input accepts integers representable exactly in float64 and binary floating values up to float64 precision. Verification interprets each stored weight as its exact rational value; arbitrary rational adjacency is outside this implementation’s input interface. Resolutions are rational. Certificate records include the graph fingerprint, block, exact resolution, verification status, and separate assembly and verification times. Isolated vertices are excluded from normalized blocks. The production implementation does not cover directed modularity, signed null models, or constrained cluster counts.

4.3 Discovery and contraction

We generate a candidate bank independently of all certificate outcomes. One Louvain discovery partition [15] at resolution 1 supplies roots. Disconnected blocks split into connected components; connected blocks split by a normalized internal Laplacian vector using original full degrees, until the minimum block size is reached. The complete bank is retained, including blocks excluded by the exact-size cap. This hierarchy proposes blocks; it provides no optimality guarantee. Originally verified blocks are united with a disjoint-set structure using Corollary 1. The degree-preserving quotient retains the discovery incumbent in the standalone certificate arm, since every proposed block refines a discovered community. A subsequent certificate stage on another method’s quotient need not preserve that representability; the sequential evaluation therefore retains the original discovery partition externally.

5 CONTRACTION FOR A METRIC SDP

The same copying argument applies to the metric semidefinite domain used for unrestricted clustering [27]. Define

$$\mathcal{M}_n = \{Y = Y^\top \geq 0 : Y_{ii} = 1, \quad 0 \leq Y_{ij} \leq 1, \\ Y_{ij} + Y_{jk} - Y_{ik} \leq 1 \quad \text{for all } i, j, k\}, \quad (12)$$

$$q_{\text{metric}}^*(A, \gamma) = \max_{Y \in \mathcal{M}_n} \langle B, Y \rangle / S. \quad (13)$$

Partition co-membership matrices are feasible, so the relaxed optimum upper-bounds the discrete optimum. The full symmetric objective includes the constant diagonal of B . We use the plain domain in Eq. (12); Hildebrand’s simplex formulation additionally requires centered PSD [26]. The bare nonnegative PSD formulation used by Wang and Kolter omits triangles [28]; its feasible domain differs from the one evaluated here.

COROLLARY 2 (METRIC FUSION). *Under the weak condition of Theorem 1, Eq. (13) has an optimum with $Y_{uv} = 1$ for all $u, v \in K$. Under the strict condition, every relaxed optimum has this property. The degree-preserving quotient therefore has the same metric optimum, with the original S , resolution and aggregated full degrees.*

Copying Gram vectors preserves PSD, unit diagonals and metric inequalities. The latter bound exterior changes by $|Y_{uw} - Y_{vw}| \leq 1 - Y_{uv}$. Degree weighting cancels the exterior null contrast, and the certificate makes the weighted mean gain nonnegative. At an optimum, each feasible anchor copy has nonpositive gain; hence every positive-probability copy preserves optimality. This proves weak fusion. The strict restricted matrix forces the block Gram matrix to be J_K . Appendix E gives the complete proof and a three-vertex counterexample for the bare PSD/box domain without triangles.

Our certificate uses a block-specific expected-improvement inequality. Objective-preserving SDP reduction through positive orthogonal projections and primal/dual affine invariance is already established [31].

The application measures the cost of a rigorously bounded relaxed optimum. Numerical SDP solutions propose a primal matrix and dual multipliers; exact rational repair and verification give $L \leq q_{\text{metric}}^* \leq U$ [29]. Separation may activate only some triangle cuts, but primal feasibility is checked against every triangle and inactive dual cuts have multiplier zero. A quotient primal lifts through the actual memberships; its upper bound transfers through the relaxation-safe stages. The lower endpoint L concerns the relaxation. A feasible partition supplies a separate discrete lower bound, so interval width alone does not prove a discrete optimum.

6 EVALUATION

6.1 Protocol and comparisons

We distinguish correctness audits, public-graph coverage, controlled mechanism studies, and isolated assembly measurements. All formal measurements use one benchmark process at a time and single-thread numerical libraries. In the discrete V1/V2 studies, candidate discovery seeds are 0,1,2; resolutions are 1/2, 1, 2; downstream Louvain seeds are 10 through 18. The complete configuration, source hashes, dataset hashes, and immutable raw outputs accompany the implementation. Language-model agents assisted research, code and drafting; Appendix D records their roles and the limits of internal agent audits.

The discrete public comparisons use six SNAP inputs: ca-GrQc, ca-HepTh, email-Eu-core, Facebook, ca-CondMat and p2p-Gnutella08 [16–20]. The processed graphs retain all vertices and have 1,005–23,133 vertices; Table 4 gives their complete size/LCC inventory and Appendix D.1 specifies preprocessing.

The scalable references use strict singleton dominance, strict fusion on positive-closure edges, compatible weak twins with proportional degrees, and strict almost-clique certification on the identical candidate bank. We also compose these criteria recursively on their current quotient, mapping only the original candidate bank to quotient identifiers. This is an explicit criterion composition rather than the entire published algorithms from which the rules come. Both total removed vertices and additional identifications beyond each reference are reported.

Controlled panels compare the same supplied block under center, full-pair, uniform and published-reference conditions. Production and forced-exact checks remain separate. Isolation enumeration gives exact evidence; bounded numerical MILP matches remain diagnostic, and native failures retain their status. Appendix D.1 specifies all oracle and solver limits.

The V1 recovery retains 139 logical jobs from 140 source attempts, including both selected native failures. Appendix D gives the recovery and analysis lineage.

V2 compares U, D, the specified recursive criteria R, and a fresh R prefix followed by a degree stage RD. RD maps the entire original bank before cap64, rechecks blocks on its actual quotient and composes the memberships. Every arm retains the original discovery partition as an external feasible fallback, selects by exact original modularity and executes all nine fixed Louvain calls, including on one-vertex quotients. The complete grid has 54 graph-discovery-resolution cases, 216 arm states and 1,944 calls; 1/3/9-call prefixes are correlated summaries. Appendix D.2 gives selection and tie handling.

Each standalone V2 workload charges discovery and its own preprocessing, quotient/interface creation, exact objective evaluation, lifting, selection and returned labels. Arm order rotates, ordinary garbage collection stays enabled, and all four states remain resident. Serialization is excluded and per-arm peak memory is not measured. Appendix D gives the disjoint accounting; V1/native times are separate.

The controlled grid includes two matched cliques, common-hub attachments, degree-heterogeneous weighted blocks, noisy planted graphs, fixed-core exterior scaling, and a private-leaf family with a proved incompatible planted block. Generator reference blocks enter only the supplied-block panel. They never seed candidate discovery. Development graphs are separate from the six public inputs and provide bounded exact-solver diagnostics.

6.2 Metric-bound application protocol

The separately frozen metric comparison contains 43 fixed whole inputs from the existing study inventory, each with at most 300 vertices: 39 controlled cases and four previously used development graphs. It uses discovery seed 0, the same complete original candidate bank in every arm, and resolution 1 except for the retained 1/8 negative control. Candidate blocks are mapped to the current quotient before applying the size cap 64. Five fresh-process arms separate the degree criterion from two relaxation-safe controls:

Arm	Preprocessing before the common metric solver
U	Original graph
S	Optimized two-copy pairs and uniform collective copies
D	Degree/profile certificates
SD	S fixed point, then a fresh D fixed point
SW	S, then fixed-degree collective full-pair PSD

The optimized pair control maximizes over constant mixtures of two anchor copies; it is distinct from full discrete Rule5. The exact fixed-degree-anchor collective contrast contains the profile bound on the same block. Different weak choices and quotients need not yield nested final pipelines. D and the full-pair stage each reach their own fixed point after S, without returning to S. Any first D acceptance after S's completed fixed point must involve at least three vertices and unequal current full degrees (Proposition 3); this necessary structure alone gives no existence or utility conclusion. Appendix H.1 gives a connected heterogeneous witness with nonzero exterior-profile dispersion: S refuses every eligible original subset, whereas D accepts the supplied K after S. Original-state discrete Rules 4 and 5 also refuse. W accepts the same K , and stronger adaptive oracle routes remain open; this instance is outside the frozen inputs and gives no measured performance claim. Nonpositive cross-component signed interactions permit a common block-diagonal component presolve in every arm. General collective-anchor optimization and general relaxed improving-map or SDP-dual verification are outside this panel. Its conclusions concern the specified criteria and sequential pipelines on these finite whole inputs and mapped blocks capped at 64.

Every arm acquires the discovery bank and its feasible incumbent inside the compute timer, including U. Models are canonicalized with CVXPY [32] and solved with SCS [33]. The common direct backend, active-triangle policy and rational denominator 2^{40} are fixed. The endpoint is a verified interval of width at most 1/1000 in the original normalized objective, within 300 compute seconds and monitored 6 GiB RSS. Complete compute charges discovery, preprocessing, quotient/component preparation, model construction, numerical solving, separation, rational repair, verification, lifting and intermediate checkpoints. Imports/input decoding and bounded final serialization are excluded; final proofs are required for credit. Thus U is a common prepared control, rather than a discovery-free solver. Appendix F specifies the numerical and exact verification policies.

The 215 case-arm tasks run once in cyclic arm order. The primary comparisons are SD/S and SD/SW at the same independently verified width target and budget. All planned outcomes remain in completion denominators; time ratios use jointly completed pairs with explicit counts. Size, family, role and resolution panels are descriptive, since parameter and seed variants are correlated. Six U-only feasibility pilots selected the common direct backend before this comparison and are excluded from its outcomes. The once-only formal study and its complete independent proof/data audit are finished; the audited descriptive outcomes follow.

6.3 Metric-bound results

All 215 planned case-arm outcomes are retained. An independent exact replay credits 182 verified-width targets and retains 33

Table 1: Fixed metric-SDP comparisons on 43 cases each. A is SD and B is the named reference. The third column gives A -only/ B -only/neither verified targets; the fourth gives faster/slower A among the 36 jointly verified pairs. Smaller median T_A/T_B favors SD. Cases share designs and are not independent samples.

Pair	Both	$A/B/N$	A faster/slower	Median T_A/T_B
SD/S	36	0/1/6	17/19	1.000379
SD/SW	36	0/0/7	15/21	1.007499

full-wall-limit outcomes as unresolved; there are no unexecuted tasks. Verified counts out of 43 are U 37, S 37, D 36, SD 36, and SW 36. The analysis uses the previously fixed complete-compute-cost ratio $T_{SD}/T_{reference}$ only when both arms meet the same exact original-normalized width target. Table 1 keeps all 43 cases in the completion denominator. Neither a wall-limited task nor its 300-second budget enters a time ratio.

Figure 1 displays the target-qualified pairs without imputing a cost to a wall-limited case. The D stage removes additional vertices after S in 12 of 43 cases and none in 31. Some controlled cases benefit against S: the best jointly verified SD/S ratio is 0.009134. Yet SW’s final quotient is no larger than SD’s in all 43 cases and strictly smaller in seven. Relative to SW, SD has no additional verified completion, no ratio below 0.5 among the 36 jointly verified cases, and seven ratios above 2; the largest is 98.857642. In that case SD uses 158 complete compute seconds versus 1.598 for SW and has 182 quotient vertices versus SW’s eight. The full ratio ranges are 0.009134–1.743446 for SD/S and 0.787373–98.857642 for SD/SW. These finite, correlated controlled/development cases do not support an IID confidence interval, broad solver claim, or a robust complete-cost advantage over the strongest implemented reference. They leave the exact contraction guarantee intact while failing to establish the planned application-level utility endpoint.

6.4 Correctness and non-dominance

An independent exact-arithmetic audit checks all 729 signed four-vertex graphs with pair weights in $\{-1, 0, 1\}$ and all 1,023 nonempty unweighted five-vertex graphs, together with 120 weighted looped cases. It checks expected-copy inequalities, fused optimal representatives, strict all-optima statements, overlap composition, sparse formulas, and quotient preservation by enumerating partitions. There are no assertion failures. These finite checks complement the proofs and do not estimate large-graph coverage.

Degree weighting and uniform weighting are incomparable. One loopless five-vertex example at $\gamma = 1/8$ has a positive uniform certificate for a two-vertex block while its exact degree-weighted pair margin is negative. We retain it as a negative mechanism control. Likewise, two matched cliques are already reducible by published collective rules; they test threshold behavior rather than separation from prior preprocessing.

Appendix B gives small degree-specific and collective examples, together with their recursive and oracle counterchecks. Proposition 1 combines both mechanisms in a connected family at $\gamma =$

$S/10^6$. Every Paley(29) block has a strict actual-median degree certificate, while every original uniform candidate and every original almost-clique subset fails, including the exhaustive polynomial search of Schulz et al. [9]. Full similar-neighborhood pairs, all repulsive cuts and all objective-faithful subgraph fusion conditions also fail. The quantified proof covers unrestricted original searches. Its growing resolution, ramp loops and surviving isolation-and-uniform composition delimit the result; it is an analytic criterion comparison separate from the frozen performance measurements.

6.5 Public coverage and downstream utility

Appendix Figure 2 reports the complete sequential comparison. Median removed fractions over three discovery seeds range from 1.69% to 35.12% for D, 0.10% to 44.41% for R, and 1.79% to 49.54% for RD. The degree stage adds removals after its own fresh recursive prefix in 44 of 54 cases, with positive median additions in 14 of 18 graph-resolution settings. The largest median addition is 1,587 vertices; the largest individual addition is 1,618. Four settings have zero median additions, including both lower resolutions of Gnutella. These are executed sequential reductions, rather than a union of original-graph weak certificates. Table 6 and Figure 4 retain every setting and its observed seed range.

Table 2 reports all five comparisons for the complete nine-call workload; Table 5 retains every correlated prefix. For each setting, speedup is the median constructed standalone T_B/T_A over the three fixed discovery seeds. Every D/U, R/U and RD/U setting has a median below one at every prefix. At nine calls, these ranges are respectively 0.555–0.753, 0.378–0.751 and 0.322–0.636. Thus none of the measured preprocessed Louvain workloads recovers its full cost against U. Adding D after R also retains overhead in every setting: nine-call RD/R medians range from 0.806 to 0.915. At that prefix, RD returns lower, equal and higher exact original modularity than R in 26, 11 and 17 of 54 seed observations. The 11 exact-quality matches have their own speedup range, 0.797–0.927; none exceeds one. Prefixes reuse the same nine calls, so their observations are not independent replications.

D is cheaper than R in 13, 12 and 11 of 18 setting medians at one, three and nine calls, respectively. This does not establish a quality-matched general advantage: at nine calls only two of 54 D/R observations have exactly equal returned modularity, with speedups 0.954 and 1.526. Safety preserves a global optimum, while a heuristic’s returned objective can change in either direction. The exact-Q comparisons therefore accompany every cost comparison; Figures 5 and 6 retain all 270 setting/comparison/prefix groups rather than selecting wins.

The separate V1 study also retains complete-cost overhead; Appendix I gives its unpooled summary and Appendix K all original observations.

The full native-solver study has 73 predeclared eligible pairs: 61 controlled and 12 development pairs. Both native arms finish in 32 pairs, with exactly equal lifted original modularity; one additional pair has an analytic one-node quotient and is reported separately. The remaining 40 pairs retain timeouts, domain exclusions or the recorded two-arm assertion. Across the 32 completed native-only pairs, median runner speedup is 1.194, but median complete

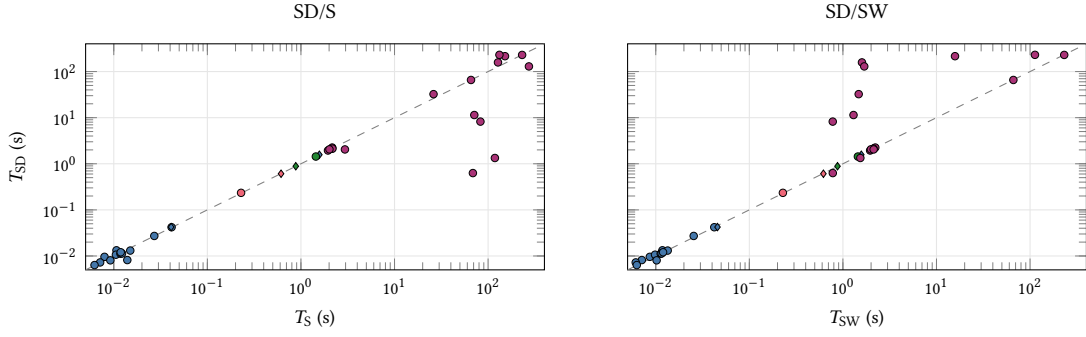


Figure 1: Audited complete-cost distribution at the fixed verified metric-SDP width target. Each panel shows all 36 jointly verified pairs among 43 planned cases; Table 1 retains other outcomes. The dashed line is equal cost, so points above it favor the reference. Blue, rose, green, and purple mark original sizes 1–32, 33–64, 65–128, and 129–300; diamonds mark development cases. Both axes use seconds on a log scale.

Table 2: Complete nine-call V2 workloads, all five comparisons. Each row retains all 18 graph/resolution settings and 54 fixed-seed cases. Speedups are standalone T_B/T_A : the range spans setting medians, and > 1 counts medians above one. Q counts are exact worse/equal/better outcomes for A versus B. Matched ranges retain every equal-Q seed and their own n . Table 5 retains all three prefixes.

Pair	k	Speedup range	$> 1/18$	Q worse/equal/better	Matches/54	Matched range (n)
D/U	9	0.55–0.75	0/18	31/3/20	3/54	0.59–0.62 (3)
R/U	9	0.38–0.75	0/18	25/2/27	2/54	0.40–0.63 (2)
RD/U	9	0.32–0.64	0/18	32/2/20	2/54	0.37–0.56 (2)
D/R	9	0.93–1.54	11/18	34/2/18	2/54	0.95–1.53 (2)
RD/R	9	0.81–0.92	0/18	26/11/17	11/54	0.80–0.93 (11)

matched pipeline speedup is 0.916; its observed range is 0.467–7.417. These ratios condition on completion and do not characterize excluded pairs. Exact lifted feasibility is independently checked; native search optimality relies on the pinned solver rather than independent exhaustive verification.

6.6 Mechanism and assembly scaling

Appendix I reports isolated assembly measurements at block sizes 16 and 32, retaining exact-checker costs and a separately labeled uncertified dense diagnostic. Median-checker time is 1.48–1.65 ms at $k = 16$ and 6.02–6.31 ms at $k = 32$, while traced peaks are 0.063 MiB and 0.209–0.240 MiB. The dense diagnostic reaches 4.70 and 8.37 MiB, respectively; it performs no exact certification and can be faster on smaller inputs. These finite measurements isolate representation cost. The current CSR slice still allocates an $O(n)$ index workspace.

Controlled supplied-block results appear in Table 9. For the nine degree-heterogeneous reference blocks at $\gamma = 1$, the median and full-pair conditions accept all nine while uniform copying accepts three. The reverse five-vertex example remains an explicit non-dominance control. On common-hub blocks at $\gamma = 1$, the median center accepts both supplied blocks while the zero center accepts one. The exploratory incompatible private-leaf blocks are rejected by every exact copy variant. Over all 17,405 recorded AUTO-versus-forced-exact block comparisons, forced exact checking accepts 64 additional blocks and there are no AUTO-only

accepts. Screening therefore loses coverage in this study; exact certification remains the acceptance gate. Unavailable uniform comparisons and numerical isolation matches retain their separate statuses and never count as certified rejections or accepts.

6.7 Exploratory global-optimum audit

A separately reviewed post-freeze audit applies the same exact proof policy to the original, degree-certificate, and recursive-reference graphs: nonpositive off-diagonal modularity weights certify the singleton optimum; otherwise, enumerate every partition only when the represented graph has at most ten vertices. Quotient proofs establish an original optimum only with their saved safety dependencies. Degree certificates are rechecked exactly; recursive dependencies replay the frozen reference implementation and do not constitute an independent reimplementations of its criteria. This audit retains 193 resolution tuples and 579 arm rows, including unresolved and unavailable rows.

Table 3 shows 59 distinct controlled tuples established in at least one arm. The degree arm establishes 32 tuples unresolved by both other arms: 22 noisy planted, eight matched-clique, and two common-hub tuples. Conversely, one tuple is established only by the recursive arm. Overall, 100 arm rows are established, 461 remain unresolved, and 18 are unavailable. All public and development tuples remain unresolved under this limited policy. Thus additional safe reductions can expose simple exact proofs on the

Table 3: Tuples with an established original optimum under the identical capped audit policy. This is exploratory proof coverage, with no timing endpoint.

Input group	Tuples	Original	Degree	Recursive
Controlled	127	15	58	27
Public	54	0	0	0
Development	12	0	0	0

controlled grid, without establishing public-graph global optima or general solver dominance.

7 CONCLUSION

Degree-weighted anchor copying cancels modularity’s exterior null model and produces a collective contraction test from sparse normalized adjacency profiles. Exact verification, compatible overlap and a quantitative loss bound accompany the certificate. On the declared public grid, it adds safe reductions after recursive criteria in 44 of 54 cases, but neither standalone nor sequential preparation amortizes in the measured Louvain workloads. Controlled quotients expose additional exact proofs under a fixed capped policy; public global optima remain unresolved. The audited metric-SDP comparison also finds no robust complete-cost advantage over the stronger full-pair control. The contribution is a complementary sufficient condition whose correctness, coverage and solver utility have separate evidence.

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A PROOF DETAILS

A.1 The exterior expansion

For fixed $w \notin K$, the initial contribution is $\sum_u B_{uw} x_u^w$. After copying to anchor r , it is $\sum_u B_{uw} x_r^w$. Hence

$$-\mathbb{E}_r \Delta F_w = \sum_u B_{uw} x_u^w - V^{-1} \sum_{u,r} d_r B_{uw} x_r^w.$$

Multiplying the first term by V/V , pairing ordered terms for $u \neq r$, and renaming r as v gives $V^{-1} \sum_{u < v} (d_u B_{uw} - d_v B_{vw}) (x_u^w - x_v^w)$. The diagonal ordered terms cancel. Summing over exterior vertices proves Eq. (3). The degree-product part of B cancels inside each bracket, proving Eq. (4). For a separated pair, $|x_u^w - x_v^w| \leq 1$; for an unseparated pair it is zero.

A.2 The center's quadratic form

Define $h_{uv} = (d_u d_v / V)(\delta_u + \delta_v)$. Expanding the pair sum for any $x \in \mathbb{R}^k$ gives

$$\begin{aligned} x^\top L(h)x &= \sum_u d_u (\delta_u + \bar{\delta}) x_u^2 \\ &\quad - \frac{2}{V} \left(\sum_u d_u \delta_u x_u \right) \left(\sum_u d_u x_u \right). \end{aligned}$$

The second term vanishes on $d_K^\perp$. Also

$$L(B_K) = L(A_K) - \frac{V}{S} (VW - d_K d_K^\top).$$

Since $P_{uv}^{\text{ext}} \leq h_{uv}$, the relaxed contrast $L(B_K - h)$ is no greater than the exact contrast in positive semidefinite order. On $d_K^\perp$ it is precisely the quadratic form in Eq. (11). Positivity there is equivalent to positivity on $1^\perp$, because both Laplacians kill 1 and $d_K^\top 1 = V > 0$. The partition-indicator decomposition then proves Theorem 1.

A.3 Quantitative loss

For each induced group K_j , translate its indicator to $y_j = 1_{K_j} - (v_j/V)1$. Then $d_K^\top y_j = 0$ and $y_j^\top W y_j = v_j - v_j^2/V$. Applying the lower Rayleigh bound to every indicator gives Eq. (8). Splitting a positive group into individual vertices decreases $\sum_j v_j^2$, so the dispersion is at most $V - \sum_u d_u^2/V$. At least one anchor attains the expected lower bound. Starting at a global optimum therefore loses at most half of $\max(0, -\mu)$ times this dispersion in F . Multiplying by $2/S$ proves Eq. (9). For disjoint blocks, copying one never splits an already fused block; summing their bounds is valid.

A.4 Center transport through coarsening

Retain the assumptions of Theorem 1. Partition the original vertices into nonempty groups, each wholly inside K or wholly outside it, and use the exact degree-preserving quotient, including its diagonal. Let C indicate the groups inside K , which form K' . For an exterior group b , transport the center as $c'_b = \sum_{w \in b} c_w$.

LEMMA 2 (CENTER TRANSPORT). *If $|K'| \geq 2$, the restricted minimum computed with the transported center satisfies $\lambda' \geq \lambda$. The resolution threshold is unchanged, so weak and strict center certificates remain valid. If $|K'| = 1$, fusion is already represented and no singleton test is needed.*

PROOF. For an internal group a , its normalized exterior row is a degree-weighted mixture of original rows followed by exterior-coordinate summation. Triangle inequality gives

$$d'_a \delta'_a \leq \sum_{u \in a} d_u \delta_u, \quad \delta' \leq \bar{\delta}.$$

Exact quotienting gives $L(A'_{K'}) = C^\top L(A_K)C$, $W' = C^\top W C$, and $S' = S$, $V' = V$. For $y \neq 0$ with $(d'_{K'})^\top y = 0$, set $x = C y$. This injective pullback has $d_K^\top x = 0$ and identical weighted norm. Because $x_u = y_a$ within group a , the deviation inequalities imply

$$\begin{aligned} y^\top [L(A'_{K'}) - \text{diag}\{d'_a(\delta'_a + \bar{\delta}')\}] y \\ \geq x^\top [L(A_K) - \text{diag}\{d_u(\delta_u + \bar{\delta})\}] x \geq \lambda y^\top W' y. \end{aligned}$$

Taking the quotient restricted minimum proves the claim. $\square$

For rational coefficients this is also an exact proof transport: the quotient threshold form is $C^\top H_\theta C$ plus a nonnegative diagonal correction, where $H_\theta = L(A_K) - \text{diag}\{d_u(\delta_u + \bar{\delta} + \theta)\}$ and $\theta = \gamma V/S$. The algebra does not require a safe coarsening; lifting a quotient optimum to an original optimum separately requires that coarsening's safety guarantee. Quotient strictness following merely weak safety does not itself imply an all-original-optima conclusion.

The center premise matters. Take nine core vertices, every distinct internal edge of weight $t = 1/16$. Six core loops have weight $1/2$; the other three loops have weight $7/16$, and each of those vertices has one edge of weight t to the same exterior vertex, which has no loop. Every core full degree is one, $V = 9$ and $S = 147/16$. The exterior profiles are six zeros and three t 's, so the original median is zero, $\bar{\delta} = t/3$, and $L(A_K) = t(9I - J)$. A contrast between two nonzero-profile vertices attains $\lambda = 23t/3 = 23/48$. Contract the six zero-profile vertices into two triples, leaving the others separate. Quotient core degrees are $(3, 3, 1, 1, 1)$; on the weighted contrast space its internal normalized Laplacian still equals $9tI$. The transported center remains zero and attains the same λ' . Its recomputed unweighted median is instead t , with deviations $(t, t, 0, 0, 0)$ and mean $2t/3$. A contrast between the two triple groups attains $\lambda'_{\text{median}} = 22t/3 = 11/24$. At $\gamma = 245/512$, the threshold is $15/32$, so the original and transported margins are $1/96$, while the new-median margin is $-1/96$. The original strict certificate fuses the entire core in every optimum, making these triple contractions safe. Thus even safe coarsening need not preserve acceptance under a recomputed median. This is an analytic example, not a measured checker run. The frozen sequential implementation rechecks its actual current quotient and median; it does not implement center transport or claim automatic acceptance containment.

A.5 Attainment of the threshold and loss bound

Take the entire loopless unit-weight complete graph on $k \geq 2$ vertices as K . Then $d_u = k - 1$, $V = S = k(k - 1)$, and the empty exterior gives $\delta = \bar{\delta} = 0$. On $1^\perp$, the normalized internal Laplacian

is $k/(k-1)$ times the identity, so $\mu = k/(k-1) - \gamma$. Every off-diagonal affinity equals $b = 1 - \gamma(k-1)/k = \mu(k-1)/k$. If $\mu > 0$, the unique optimum fuses all vertices; at $\mu = 0$, every partition is optimal; if $\mu < 0$, the unique optimum consists of singletons. Thus weak fusion at equality and the strict all-optima statement have the sharp distinction in Theorem 1. In the negative-margin case, singleton and fused modularities are $-\gamma/k$ and $1 - \gamma$, respectively. Their gap is

$$\gamma(1 - 1/k) - 1 = -\mu(1 - 1/k),$$

exactly Eq. (9), since $\sum_u d_u^2/V = k - 1$. For any partition with block sizes h_j , its expected copy gain is $b(k^2 - \sum_j h_j^2)/2$, also exactly Eq. (8). This elementary attainment example concerns the bounds' worst case, not practical completeness of the sufficient certificate.

A.6 Overlap and the zero-degree exception

Within a connected component of the block-intersection graph, choose a traversal in which each new block intersects the previous union. Apply the first certificate to an optimum. At every subsequent step, select an anchor in the intersection. It has positive degree and hence positive probability. The optimal-anchor lemma preserves the objective, and all vertices in the previous union remain in its community. Disjoint intersection components can be processed successively.

The positive-degree requirement is substantive. Take unit edges (1, 4) and (2, 3) and isolated vertex 0, at resolution 1. Each edge plus vertex 0 can be fused in some optimum, but the optimum separates the two edges. Requiring both blocks fused forces all positive-degree vertices together. An anchor at vertex 0 has zero probability, so the overlap proof cannot use that intersection.

B CRITERION COMPARISONS

B.1 The pair specialization

For a separated attractive pair, write $a(P)$ and $b(P)$ for the exterior losses of its two directional copies, excluding the mutual gain B_{uv} . For any positive anchor probabilities, the smaller loss satisfies

$$\min\{a(P), b(P)\} \leq \pi_v a(P) + \pi_u b(P) \leq \sum_{w \notin \{u,v\}} |\pi_v B_{uw} - \pi_u B_{vw}|.$$

The last inequality bounds the expected exterior loss. Maximizing the left side over exterior assignments gives the exact similar-neighborhood threshold of Böcker et al. [8]. Thus acceptance by the degree-copy pair penalty implies acceptance by that rule. A center relaxation can only increase the penalty. Zero mutual affinity with zero penalty is covered by proportional weighted twins [10]. We do not attribute a new strongest pair rule to the proposed certificate.

B.2 A degree-specific collective example

The following rational construction is an algebraic comparison, obtained during the post-freeze conceptual review; it is not a benchmark point. Let $K = \{0, 1, 2, 3\}$ have cycle weights $A_{01} = 8$ and $A_{12} = A_{23} = A_{30} = 6$. There are 20 independent exterior vertices, each adjacent to vertices 0,1 with weight 7 and to vertices 2,3 with weight 6. Divide these exterior vertices into two classes of ten, and attach each class to its own new anchor by edges of weight 200. All

other entries, including loops, are zero. Use $\gamma = 2273/1750$. Then

$$\begin{aligned} d_K &= (154, 154, 132, 132), & d_E &= 226, \\ d_{\text{anchor}} &= 2000, & S &= 9092, \\ \gamma/S &= 1/7000. \end{aligned}$$

The normalized exterior profiles equal $1/22$ on each exterior vertex and zero on the anchors. Hence a median center gives $\delta = 0$. The restricted internal spectrum is $\{13/154, 15/154, 2/11\}$. To obtain it, divide the cycle weights by two: its internal degrees are (7, 7, 6, 6), and the normalized adjacency has bipartite singular values 1 and $1/14$. The original full degrees are 22 times these degrees. Consequently

$$\lambda - \gamma V/S = 13/154 - 143/1750 = 26/9625 > 0.$$

The proposed test therefore forces all of K together in every optimum.

For a high-degree and a low-degree core vertex, the uniform exterior row contrast is $D = 20(507/1750) + 2(44/7) = 3214/175$. The signed internal cut between $\{0, 1\}$ and $\{2, 3\}$ is $48/125$. After uniform-copy penalties, its energy is $48/125 - D = -15734/875 < 0$, proving failure of the uniform PSD test.

For direct positive-closure comparison, retain every internal signed affinity of an induced witness H , as in our reference implementation of Lange's Appendix Theorem 4 [7]. Write $b_H = 3549/1750$ and $b_L = 1521/875$ for the core-to-exterior affinities. Any witness containing an anchor has a negative cut separating its core vertices: each high-degree vertex contributes at most $20b_H - 44 = -602/175$, and each low-degree vertex at most $20b_L - 264/7 = -516/175$. Its zero minimum-multicut premise fails. If a witness contains exterior vertices but no anchor, its positive boundary is at least $948/7$, while a singleton core cut in its positive closure costs at most $13489/1750 + 20b_H = 84469/1750 < 948/7$. A witness containing only core vertices and a target pair has boundary at least $40b_L = 60840/875$, also larger than this cut bound. Thus these direct full-induced witnesses cannot certify a core pair. The comparison does not assert separation under a selected-edge convention that discards internal objective interactions. Isolating $C = K$ also fails the C-subinstance premise: adding one exterior vertex gains $2b_H + 2b_L = 6591/875 > 0$.

B.3 The recursive countercheck

The same construction has a stronger prior reduction sequence. An exterior class with its anchor has internal cut $q[s - (10 - q)c] > 0$ when q class vertices are separated, where $s = 948/7$ and $c = 12769/1750$. Its positive boundary is $10T$, where $T = 6591/875$. Every target-separating positive-closure cut costs at least

$$\begin{aligned} \min_{1 \leq q \leq 10} \{qs - q(10 - q)c + \min(q, 10 - q)T\} \\ = 135261/1750 > 10T = 131820/1750. \end{aligned}$$

Hence prior positive-closure conditions strictly fuse each exterior class and its anchor. After both fusions, core-to-class affinities are negative: $-4151/175$ for high-degree vertices and $-3558/175$ for low-degree vertices. The almost-clique condition on K then has positive internal minimum cut $5418/875$, negative internal mass $5082/875$, and zero positive exterior mass. It certifies K strictly by

336/875. The new condition therefore provides a direct local certificate in this example, while a stronger recursive collective route recovers the same fusion. This distinction must not be reported as superiority over complete prior preprocessing or exact optimization.

B.4 Collective fusion beyond the cheap starting fixed point

Let P be the Paley graph on $\mathbb{F}_{29}$: two distinct residues are adjacent when their difference is a nonzero square. It is undirected and 14-regular. The quadratic-character sum $\sum_x \chi((x-a)(x-b)) = -1$ for $a \neq b$ gives six common neighbors for adjacent vertices and seven for nonadjacent vertices. To verify the sum, an affine change reduces it to $\sum_x \chi(x^2 - c) = -1$ for $c \neq 0$. The equation $y^2 = x^2 - c$ has 28 solutions, since $(x-y)(x+y) = c$ and 2 is invertible. These counts give

$$A_P^2 + A_P = 7(I + J), \quad \lambda_2(L_P) = (29 - \sqrt{29})/2.$$

The two nonconstant adjacency eigenvalues are $(-1 \pm \sqrt{29})/2$, each with multiplicity 14.

Connect two copies of P by their corresponding-vertex perfect matching. The resulting graph G has 58 vertices, 435 edges, full degree 15 and $S = 870$. At $\gamma = 1$, every positive pair affinity is $\alpha = 43/58$ and every negative one is $-\beta = -15/58$. For a whole copy K , the median exterior center is zero and $\delta_u = \bar{\delta} = 1/15$: each exterior column has one matching incidence and 28 zeros. Its block volume is $V = 435$. Theorem 1 has margin

$$\mu = \frac{\lambda_2(L_P) - 2}{15} - \frac{1}{2} = \frac{10 - \sqrt{29}}{30} > 0.$$

Thus both copies fuse in every optimum. Their quotient has

$$A' = \begin{pmatrix} 406 & 29 \\ 29 & 406 \end{pmatrix}, \quad d' = (435, 435), \quad B'_{12} = -377/2.$$

Its separated partition has $Q = 13/30$; joining gives zero. The two-copy partition is consequently the unique original optimum, up to labels. An independent rational projection of $L_P - (19/2)I$ onto $1^\perp$ has 28 strictly positive exact elimination pivots. It verifies the same definiteness without floating eigenvalues; the second copy follows by isomorphism. This is a certificate-plus-quotient proof, not enumeration of all 58-vertex partitions.

No original subset passes almost-clique. For an arbitrary subset U of size $h \geq 2$, let e, m, b count its positive internal edges, internal nonedges and positive boundary edges. Write κ for its positive weighted internal mincut, with $\kappa = 0$ if disconnected. Böcker Rule4 [8] requires $\kappa \geq \beta m + \alpha b$, while a singleton cut always gives $\kappa \leq \alpha \min(h-1, 15)$. For $2 \leq h \leq 15$, regularity gives $b = 15h - 2e \geq h(16-h) > h-1$, so boundary mass alone defeats the condition. The symmetric and antisymmetric modes of $\begin{pmatrix} A_P & I \\ I & A_P \end{pmatrix}$ give $\lambda_2(L_G) = 2$. For $16 \leq h \leq 31$, the centered indicator bound gives $b \geq 2h(58-h)/58 \geq 1344/58 > 15$, again exceeding the mincut bound. For $32 \leq h \leq 58$, $e \leq 15h/2$ gives $m \geq h(h-16)/2 \geq 256$, and $\beta m \geq 3840/58 > 645/58 = 15\alpha$. These exhaustive ranges exclude even the weak condition on every original subset, independently of the proposed bank.

No eligible original pair passes similar-neighborhood. Check the full max-min expression of Böcker Rule5 [8], rather than replacing it by a general half-contrast bound. A positive edge has $t = 6$ common positive neighbors inside a copy and $t = 0$ across the matching. Both exclusive-neighborhood offsets in the source rule are $a = 14 - t$. On its remaining set W , the signed rows agree, so the two directional terms are $a+z$ and $a-z$ for any two disjoint chosen subsets of W . Their minimum is at most a , and the empty subsets attain a . The exact right-hand side is therefore 8 or 14, both greater than $B_{uv} = 43/58$. Negative pairs are ineligible for this attractive rule.

The specified recursion and the oracle boundary. Singleton dominance has margin $-1189/58$ on every positive edge. The single-edge positive-closure criterion [7] has margin $(t-27)\alpha < 0$. Every positive edge has an exclusive positive neighbor, at which its two signed rows have opposite signs; no positive proportional factor satisfies Twin Simple [10]. Together with the all-subset almost-clique failure, these facts show that the specified cheap pure-merge recursion has no first contraction. Adding original Rule5 would not supply one. Arbitrary-subgraph criteria, other cannot-link or bound-based reductions, and complete solving are outside this exclusion.

Uniform copying also yields $L_P - (19/2)I$ on $1^\perp$; equal degrees make this example unable to distinguish the anchor distributions. Moreover, the supplied-C isolation premise [10] holds for $C = K$. The C-subinstance retains all interactions incident to K and zeros only exterior-exterior interactions. Its same uniform-copy certificate strictly fuses K . Each exterior vertex has total affinity $\alpha - 28\beta = -377/58 < 0$ to fused K ; removing it incurs no exterior-exterior loss. Isolating K is therefore subinstance-optimal. Scaling all signed costs by 58 gives equivalent integer costs 43 and -15 , consistent with the source's integer formulation. The example distinguishes a cheap sufficient certificate from the named local merge rules, while remaining covered by the more general isolation oracle.

B.5 A joint original-instance separation

The following construction distinguishes degree-weighted collectivity from unrestricted searches of several named initial predicates. All costs in this subsection use the original $B = A - \beta dd^T$ with $\beta = 10^{-6}$. The positive-support graph is connected.

For an integer $m \geq 5000$, take two core Paley(29) copies P_1, P_2 with weight 999 on Paley edges and their same-index matching. Each has a loop 999 at index 0 and an additional loop $39i$ at index $i = 0, \dots, 28$. For each $j \in \mathbb{Z}_m$, take remote copies R_j^a, R_j^b with the same weight 999 Paley edges and matching, and loops $39i$, without the extra index 0 loop. Add weight 400 edges from $(R_j^b)_i$ to $(R_{j+1}^a)_i$. A bridge w_j connects to $(P_1)_0$ with weight $36/m$, to each other P_1 vertex with weight $35/m$, and to $(R_j^a)_0$ with weight $600/m$. Set $\gamma = \beta S$. Scaling the whole adjacency by m , while keeping γ fixed, gives an equivalent integer-weight instance.

The core volumes are $V_1 = 452414$, $V_2 = 451398$; remote volumes are $V_R = 461999$ for each b -half and $V_R + \epsilon$ for each a -half, where $\epsilon = 600/m$. Bridges have degree $1616/m$, and

$$n = 58 + 59m, \quad S = 906028 + 923998m.$$

Core degrees are 16020 at $(P_1)_0$ and $15020 + 39i$ otherwise; they are 15984 at $(P_2)_0$ and $14985 + 39i$ otherwise. Remote degrees are $15385 + 39i$, with extra ϵ at $(R_j^a)_0$. Every nonbridge degree lies in $[15024, 16477]$ and every off-diagonal adjacency row sums to less than 15386 .

PROPOSITION 1. *For every integer $m \geq 5000$, all $2m + 2$ Paley copies have strict degree-profile certificates with their actual coordinate medians. The unique optimal partition consists of P_1 with all bridges, P_2 separately, and every remote Paley copy separately. No nontrivial original subset passes the uniform-copy PSD condition or almost-clique Rule 4. No original positive pair passes full Rule 5, singleton dominance, single-edge positive closure or positive-factor twins; original heavy Rules 1–3 also refuse. No original negative pair passes Lange’s repulsive-cut condition for any separating cut, and no target passes its subgraph-fusion condition on an objective-faithful original subgraph. A conditional isolation followed by uniform copying nevertheless recovers the same private and core fusions.*

Degree certificates and the unique partition. The Paley gap exceeds $23/2$. Weighted variance transfers its Poincaré bound to the full-degree normalization, giving $\lambda \geq 999(23/2)/d_{\max}$. The actual P_1 bridge-column median is $1/(445m)$: the neighboring crossproducts are 324 and 1080 , placing fourteen ratios on either side of index 0 . Thus

$$\delta_{\max,1} = \frac{40461}{609205}, \quad \delta_1 = \frac{12899739}{201324230}.$$

With $d_{\max,1} = 16112$, the resulting margin exceeds 0.1301 . P_2 has zero exterior medians, with $\delta_{\max,2} \leq 999/14985$, $\delta_2 = 28971/V_2$, $d_{\max,2} = 16077$, and margin greater than 0.132 . Remote medians are also zero. Using $\delta_{\max} \leq (1399+3/25)/15385$, $\delta \leq (40571+3/25)/V_R$, $d_{\max} = 16477$ and threshold $\beta(V_R+3/25)$ gives margin greater than 0.0564 . These bounds cover all m in the domain.

Strict fusion leaves every remote block with negative aggregate affinity to every exterior vertex; even its matching, cycle and own bridge incidences are negative. The same holds for P_2 after core fusion. The remaining positive affinities are $B_{P_1, w_j} = \alpha/m$, where $\alpha = 1016 - \beta V_1 1616 = 8903093/31250$. Bridge pairs have affinity $-\beta 1616^2/m^2$. Since $\alpha - \beta 1616^2 > 0$, each remaining bridge strictly prefers P_1 , proving uniqueness. Its modularity is

$$T_m = 454030^2 + 451398^2 + m(V_R + \epsilon)^2 + mV_R^2, \\ Q^* = \frac{846886 + 842856m - \beta T_m}{906028 + 923998m}.$$

Every uniform candidate. Write $D_{uv} = \sum_{w \notin U} |B_{uw} - B_{vw}|$. A passing $L(B_U - D/|U|) \geq 0$ would have an optimal fused representative, so uniqueness first restricts U to one optimal class. For $2 \leq |U| \leq 64$, a subset without an observed internal edge has a negative singleton cut. Otherwise its degree range is at least 14 , so one endpoint has average absolute degree distance at least 7 . At most 64 remote matched components have observed incidence from U ; the untouched mass is at least $923998(m - 64)$. Its uniform singleton penalty is therefore at least $7\beta 923998(m - 64) > 15386$, above the internal signed cut bound. Larger candidates must mix P_1 vertices and bridges, or consist of bridges alone. The latter have negative cuts. For a selected core u and bridge v , remote adjacency

reinforces the null contrast:

$$D_{uv} \geq \beta(d_u - 1616/m) 923998m + 600/m.$$

Their bridge singleton penalty is at least $\beta 923998(15020m - 1616)/(m + 29) > 1016/m$, its largest possible internal signed cut. Thus every nontrivial original uniform subset fails.

Every almost-clique subset. Set $a = 999 - \beta 16477^2$, $A = 999 - \beta 15024^2$, $c = 400 - \beta 16477^2$, $C = 400 - \beta 15024^2$, and $b = \beta 15024^2$. The old weight 999 components are matched 58 graphs, 15 -regular with Laplacian gap 2 . For a bridge-free subset containing $s \geq 2$ vertices of one such component, its positive mincut is at most $(s - 1)A + C$, or $15A + C$ when $s \geq 15$. The Rule 4 right side has the lower bounds

$$\begin{cases} s(16 - s)a, & 2 \leq s \leq 14, \\ (645/29)a, & 15 \leq s \leq 31, \\ 256b, & 32 \leq s \leq 58. \end{cases}$$

Each exceeds the corresponding mincut upper bound. In the first range the difference is concave and both endpoints are positive; the other gaps exceed 4407 and 46010 . If at most one vertex is selected per old component, internal positive edges form a cycle matching with mincut at most C , while the old exterior boundary is at least $15a > C$. A subset containing a bridge has mincut at most $1616/m$. A proper old-component part has boundary at least a ; a whole selected component has negative internal mass at least b . Both exceed this mincut. Bridge-only subsets have positive deficit and zero mincut. This proves refusal for every original subset, including the exhaustive polynomial-time Rule 4 search [9].

Full pair and initial local conditions. Use the exact Rule 5 maximum over disjoint C_u, C_v in its remaining set W , with exclusive-neighborhood offsets Δ_u, Δ_v as defined by Böcker et al. [8]. The empty assignment lower-bounds both directional thresholds by $8a$ for a Paley edge, $14a$ for a matching edge and $15a$ for a cycle edge. These exceed the respective target weights. For a P_1 -bridge edge, the core has its exclusive matching neighbor and the bridge has its exclusive remote neighbor; both offsets exceed $36/m$. For $u = (R_j^a)_0, v = w_j$, choose $C_u = R_k^a$ for any $k \neq j$ and $C_v = \emptyset$. This is an admissible subset of W . Its two terms satisfy

$$\Delta_v - B(v, C_u) \geq \frac{600}{m} + \frac{5393617}{12500m}, \\ \Delta_u + B(u, C_u) \geq \frac{42859700447}{12500000}.$$

Both exceed $600/m$, covering the delicate remote bridge edge.

Every nonbridge has positive signed row mass at least $15a > 3 \cdot 999$. This defeats heavy Rules 2–3, singleton dominance and Eq 21 positive closure; the latter’s repayment is at most the other endpoint’s positive row mass. A bridge’s core positive row exceeds $72/m$. For its remote target, its own remote edge plus the negative incidences to another a -half give absolute row mass greater than $1200/m$. Each positive pair also has an exterior neighbor with opposed signs in its two signed rows, excluding every positive proportional factor in Twin Simple [10]. Rule 1 is covered by the repulsive-cut proof below.

Every repulsive cut. Lange’s Eq.(3) requires $|B_f|$ to dominate the positive capacity of a separating cut [7]. A cut splitting an old matched 58 component crosses at least 15 old edges, costing $15a$. If all old components are whole but remote side labels differ, two cycle transitions cost at least $58c$. If the core and the entire remote ring lie on opposite sides, each bridge path costs at least α/m , because $600 - \beta 1616(15385 + \epsilon) > \alpha$. Thus every cut separating nonbridge vertices costs at least $\min(15a, 58c, \alpha) = \alpha > \beta 16477^2$, the maximum negative pair magnitude. For a bridge target, those cases already suffice; otherwise all nonbridges share a side and a separated bridge costs at least α/m . This exceeds both $\beta 1616 16477/m$ and $\beta 1616^2/m^2$. All separating cuts are covered, including the source’s connected-cut restriction.

Every objective-faithful subgraph. For Lange’s Eq.(19), retain every negative internal objective term of H ; positive terms may initially be omitted. First take the full signed induced H and suppose its minimum multicut cost is zero, as required by the source. For $s > 0$ selected vertices in one Paley copy and t selected nonbridges outside it, their internal observed cut is at most $1415s$, while the null cut is at least $\beta 15024^2 st$. Nonnegative internal cuts imply $t \leq 6$. Hence H meeting two copies has at most 12 nonbridges, and otherwise at most 29. With at least two nonbridges, the old matched 58 boundary is at least $645/29$ edges: use $s(16-s) \geq 28$ for $2 \leq s \leq 14$, the gap bound $s(58-s)/29 \geq 645/29$ for $15 \leq s \leq 29$, or two singleton components contributing 30 edges. Its positive exterior cost exceeds $(645/29)a > 15386$. The singleton positive-closure cut at any target endpoint is at most 15386 for a nonbridge and $1616/m$ for a bridge, so Eq.(19) fails. With one nonbridge the boundary is at least $15a$, while every target has a bridge endpoint. With no nonbridges a nontrivial H has negative multicut cost, violating the premise. Finally, adding omitted positive internal terms preserves the zero minimum, increases all closure cuts and leaves the exterior cost fixed. A passing subgraph retaining all negative terms would therefore imply a passing induced one, already excluded.

A surviving recursive composition. Let $T_j = R_j^a \cup R_j^b$ and let r count singleton intersections of the current communities with T_j . An exterior cycle vertex joined to at least two vertices of T_j has aggregate affinity at most $400 - 2\beta 15385^2 < 0$. Isolating T_j therefore loses at most $rc_{\max} + 600/m$, with $c_{\max} = 400 - \beta 15385^2$. Inside the isolated component, uniform copying either half has restricted signed floor at least

$$h = 999(23/2) - 1998 - \beta(V_R + 3/25)(16477 + 1092) = \frac{17170468259}{12500000}.$$

Original degrees and β remain fixed. The two successive copies have expected gain at least $14hr/29$ when $r \geq 1$, or $28h/29$ when $r = 0$ and either half splits. Copying one half leaves the other half’s internal relations unchanged. Subtracting isolation cost leaves strictly positive lower bounds

$$\frac{362292324377}{725000000}, \quad \frac{120182402813}{90625000}.$$

Take identity if neither half splits; otherwise some anchor pair improves. This establishes the private fusions through a composed improving map. Their negative exterior aggregates permit exact removal; the residual uniform core floors exceed $172318189/100000$ and $438820191/250000$. The SEA incident-subinstance isolation

Table 4: Public graph inputs. Counts are preprocessing meta-data, not performance results. LCC denotes the largest connected component.

Graph	Vertices	Edges	LCC
ca-GrQc	5,242	14,484	4,158
ca-HepTh	9,877	25,973	8,638
email-Eu-core	1,005	16,064	986
Facebook	4,039	88,234	4,039
ca-CondMat	23,133	93,439	21,363
p2p-Gnutella08	6,301	20,777	6,299

oracle also covers each private half, with positive closed neighborhoods of 88 and 87 vertices. These are analytic counterexamples, rather than measurements of a released implementation. The proposition concerns initial predicates at a resolution growing with m ; recursive cut-and-copy optimization remains capable of recovering the same optimum.

C IMPLEMENTATION BOUNDARIES

An accepted certificate refers to one immutable input graph and one exact resolution. Certificate composition rejects stale graph fingerprints and different resolutions. Quotient assembly accumulates rationally representable weights and validates degree and total-volume preservation. The sparse core contains no approximate automatic merge policy.

The exact verifier uses a block-size cap 64. The production floating screen has tolerance 10^{-10} and abstains when the margin is not comfortably positive. Separate forced-exact controlled panels expose boundary cases without changing the production acceptance count. The uniform reference uses an explicit 150-vertex graph and exterior cap; the signed closure oracle enumerates at most 16 internal vertices. Domain exclusions are reported with their reason.

D MEASUREMENT LINEAGE AND ADDITIONAL CONTEXT

D.1 Discrete inputs and detailed comparison controls

Table 4 lists six public SNAP inputs [16–20] after the declared preprocessing: remove self-loops, take the simple undirected union, retain every listed endpoint, and relabel sorted original vertex identifiers. Directed sources use the same union policy. Reported edge counts refer to these processed graphs.

Controlled studies compare the same supplied block under the median, zero, anchor, full-pair, uniform-copy, positive-closure, and almost-clique conditions. Forced exact threshold checks are recorded separately from the production floating-screened checker. The C-subinstance oracle enumerates all partitions up to 10 active vertices, and uses a bounded numerical MILP up to 50 otherwise. Enumeration optimum equality is an exact certificate; numerical MILP matches remain diagnostics without an exact optimum proof. Larger and timed-out cases remain unavailable or unresolved. Weighted native KaPoCE preprocessing and its full exact

solver operate under explicit input and integer-domain limits; their original source is pinned and overflow failures are recorded.

D.2 Public V2 composition and selection

The separately frozen public-pipeline extension regenerates the same complete banks and compares four arms: unreduced discovery-incumbent control (U), the degree certificate alone (D), recursive criteria alone (R), and a fresh R prefix followed by one degree stage (RD). RD maps the entire original bank, including originally oversized blocks, through its actual quotient before applying the cap of 64. It verifies new certificates on that quotient and composes the actual memberships; it does not union unrelated original-graph weak certificates. Each arm retains the original discovery partition as an external feasible fallback, whether its quotient can represent that partition or not. Exact original modularity selects the returned solution, with ties favoring discovery and then the earliest downstream seed. All nine Louvain calls execute unchanged in every arm, including a one-vertex quotient. The completed grid has 54 graph-discovery-resolution cases, 216 arm states and 1,944 calls. The one-, three- and nine-call prefixes are correlated summaries of those calls.

D.3 Discrete experiment lineage

The first frozen run halted on a native assertion in the Davis development case. A separately reviewed recovery reran that entire case and all previously unlaunched scaling jobs under a fixed selection policy. The resulting evidence contains 139 logical jobs from 140 source attempts; both selected native failures remain failed, with no optimum, bound or completed-time inference. An independently reviewed analysis-only repair validates producer-resolved generator metadata without changing measurements, endpoints or selection. Original run records, incident evidence and all revisions remain archived.

V2 times are constructed standalone costs at one resolution. Regenerated setup cost is charged once to each such workload even when physically shared across resolutions. Discovery is charged in every arm; bank refinement is charged to D, R and RD. To avoid overlap in the unchanged candidate routine, its outer duration B is partitioned into $B - F$ and its measured refinement duration F ; the required identity validations are charged in disjoint phases. Each R and RD arm pays for its own recursive prefix. Certification, exact degree-cache construction, quotient/interface creation, objective evaluation, lifting, selection and returned-label materialization are included. Arm order rotates, ordinary garbage collection stays enabled, and all four states remain resident within a case. Input/output serialization is excluded and per-arm peak memory is not measured. V1/native timings are not pooled with V2.

Research-tool provenance. Language-model agents in Codex assisted literature discovery, candidate mathematics, implementation, analysis, manuscript drafting and internal reviews. Exact arithmetic checks and separately assigned agent audits retain explicit claim scopes and immutable evidence records. These internal reviews are not human peer review; no human signoff is inferred from their verdicts. The manuscript and reproducibility record disclose the resulting checks, failures and unresolved claims.

D.4 Other graph certificates

Other graph certificates concern different objects. Randomized smoothing certifies a detector’s smoothed co-membership predicate against bounded edge edits [21]; sampling-model analyses give confidence sets for population eigenspaces and, under additional assumptions, labels [22]. Empirical spectral stability also supports private release and model-based recovery [23].

D.5 Additional optimization and spectral comparisons

Parametric clustering stability [12] and low-rank correlation clustering [13] address related global optimization questions. Here the object of certification is a supplied local block, whose internal matrix may coexist with an arbitrary exterior graph. Collective join and cut criteria also extend to cubic correlation clustering on general graphs [11]. Our pairwise modularity model retains the original degree-null interaction and studies a supplied block’s fusion.

Normalized modularity spectral analysis relates resolution to the graph Laplacian’s Fiedler space [14]. The present local test accounts for exterior assignments and guarantees a discrete optimal fused representative. Internal modularity-block Laplacian PSD structure also appears in convexified modularity recovery [25]: its dual construction contains $L(A_K) - \lambda(VW - d_K d_K^T)$, which is $L(B_K)$ when $\lambda = \gamma/S$. That proof controls a complete planted partition and model-based cross-block interactions. Our local condition instead bounds observed exterior assignments explicitly, without a sampling model or a supplied complete partition. When the block contains the entire graph, the zero-exterior case recovers the established normalized-Laplacian resolution threshold. For an isolated connected component at $\gamma = 1$, Louf et al. give an exact nonsplitting threshold in terms of a normalized cut minimum [1]. The restricted normalized-Laplacian eigenvalue is no larger than that cut minimum, so our zero-exterior condition is a sufficient spectral relaxation of their component criterion. Our supplied blocks may instead have arbitrary exterior connections, whose normalized profiles enter the certificate.

Metric semidefinite relaxations of unrestricted clustering combine PSD, unit diagonals, nonnegative co-memberships and all triangle inequalities [27]. For modularity, simplex-Gram coordinates yield an established metric SDP with an additional centered PSD constraint [26]. The bare nonnegative PSD formulation and its low-cardinality factor optimization use a different domain without triangles [28]. Our condition supplies an optimal fused representative on the metric domain, and strict fusion in every relaxed optimum; Appendix E proves both statements, including preservation of the optional centered constraint. Its exact three-vertex counterexample shows why triangles are needed. Rigorous postprocessing of numerical SDP proposals is also established [29]. It provides a way to verify objective bounds when evaluating contraction for this relaxation.

E A METRIC AND SEMIDEFINITE INTERPRETATION

This appendix interprets the same sufficient condition within established relaxed improving-map and metric SDP frameworks [26, 27, 30]. It introduces no additional production criterion or global

integrality guarantee. Define the off-diagonal pair penalty $h_{uv} = d_u d_v (\delta_u + \delta_v)/V$ and the signed pair Laplacian $H = L(B_K - h)$. Appendix A.2 and translation along **1** show that Theorem 1's weak condition is equivalent to $H \geq 0$; its strict condition gives $\ker H = \text{span}\{\mathbf{1}\}$.

For n original vertices, let $J_n = \mathbf{1}\mathbf{1}^\top$ and consider

$$\mathcal{R} = \{Y = Y^\top \geq 0 : Y_{ii} = 1, \quad 0 \leq Y_{ij} \leq 1, \\ Y_{ij} + Y_{jk} - Y_{ik} \leq 1 \quad \text{for all } i, j, k\}.$$

Every unrestricted partition co-membership matrix belongs to this compact convex set. One may additionally impose $Y - J_n/n \geq 0$; the claims below hold for both choices. No cardinality, balance or prescribed cluster-size constraint is imposed.

Let $f_r(u) = r$ for $u \in K$ and $f_r(u) = u$ otherwise. With selector matrix $(R_r)_{u,:} = e_{f_r(u)}^\top$, define $T_r(Y) = R_r Y R_r^\top$ and $T(Y) = \sum_{r \in K} (d_r/V) T_r(Y)$. Index substitution preserves bounds, diagonals and metric inequalities; congruence preserves PSD. Since $R_r \mathbf{1} = \mathbf{1}$, it also preserves the optional centered PSD constraint. Convexity makes $T(Y)$ feasible, and every copied matrix has its $K \times K$ block equal to J_K .

Write $F(Y) = \sum_{u < v} B_{uv} Y_{uv}$. The paired exterior expansion of Eq. (3) remains linear in Y . Metric inequalities give $|Y_{uw} - Y_{vw}| \leq 1 - Y_{uv}$ for $u, v \in K, w \notin K$. Together with the degree-null cancellation and center bound, this yields

$$F(T(Y)) - F(Y) \geq \sum_{u < v \in K} (B_{uv} - h_{uv})(1 - Y_{uv}) \\ = \frac{1}{2} \text{tr}(H Y_{KK}) \geq 0.$$

The trace identity uses unit diagonals; the final inequality uses PSD of both factors.

At a relaxed optimum, each feasible T_r has gain at most zero, while their positive weighted mean has gain at least zero. Every anchor therefore preserves optimality and gives an optimal fused representative. Under the strict condition, the same optimality sandwich forces zero trace, so $\text{range}(Y_{KK}) \subseteq \ker H$. Unit diagonals then force $Y_{KK} = J_K$: every relaxed optimum fuses the block. The average T is generally fractional; an individual anchor need not improve an arbitrary nonoptimal input.

Hildebrand's simplex-Gram coordinates translate as $Y = (J_n + (p - 1)X)/p$, with $p = n$ for unrestricted partitions. The literal translation includes $Y - J_n/p \geq 0$, rather than plain PSD alone; the objective differs by a fixed affine constant. The argument above preserves that constraint. For a membership matrix C with nonempty groups and $C \mathbf{1}_m = \mathbf{1}_n$, its quotient transport is

$$Y - J_n/n = C(Z - J_m/n)C^\top.$$

Principal representatives prove the reverse PSD implication. Exact transport therefore retains the original coefficient $1/n$ through successive contractions; replacing it by $1/m$ is not generally equivalent. This observation concerns the optional centered constraint. The production solver and signed-component presolve use the plain metric domain. CMM's base relaxation [25] omits the metric inequalities, so this proof does not establish relaxed improvement for that bare feasible set. It also does not claim an earlier identical observed-profile formula or a complete dual-variable construction.

E.1 The metric premise is essential

Consider three vertices u, v, w with $A_{uv} = 10, A_{vw} = 1, A_{uu} = 1$ and all other entries zero apart from symmetry. At $\gamma = 1, d = (11, 11, 1), S = 23$ and $V = 22$ for $K = \{u, v\}$. Both the zero and median centers give $\lambda = 19/11$, with strict margin $195/253$. Write $a = B_{uv} = 109/23, b = B_{uw} = -11/23, c = B_{vw} = 12/23$. The unique discrete optimum fuses all three: $F = 110/23$ and $Q = 0$.

Every unit-diagonal PSD/box matrix with $Y_{uv} = 1$ has identical u, v Gram vectors. Hence $Y_{uw} = Y_{vw} = z \in [0, 1]$ and $F(Y) = a + (b + c)z \leq 110/23$. For $\varepsilon = 1/1000$, let $x = 1 - \varepsilon, h = \sqrt{1 + 264\varepsilon}$, and choose unit vectors p_u, p_v with $p_u^\top p_v = x$. Set $p_w = (12p_v - 11p_u)/h$. Its norm is one, and their Gram matrix X has

$$X_{uv} = x, \quad X_{uw} = (1 - 12\varepsilon)/h, \quad X_{vw} = (1 + 11\varepsilon)/h.$$

These entries are in $[0, 1]$. Thus $Y = J_3/3 + (2/3)X$ is also PSD/box feasible, including $Y - J_3/3 \geq 0$. Its objective satisfies

$$F(Y) - 110/23 = \frac{2(h - 1 - 109\varepsilon)}{69} > 0,$$

because $h^2 - (1 + 109\varepsilon)^2 = 46\varepsilon - 11881\varepsilon^2 > 0$. Both bare PSD/box domains therefore have every optimum $Y_{uv} < 1$, despite the strict discrete certificate. This means the pair is unfused, not that $Y_{uv} = 0$ or that the optimum is integral. The witness violates the metric inequality by

$$Y_{uv} + Y_{vw} - Y_{uw} - 1 = \frac{2}{3}\varepsilon(23/h - 1) > 0.$$

It is excluded by the stated metric feasible set. This analytic example shows that its premise is necessary for extending this certificate to these relaxed domains; it makes no solver-performance or all-SDP claim.

F METRIC-BOUND PROTOCOL AND VERIFICATION

F.1 Relaxation-safe comparison criteria

The comparison criteria use the same feasible anchor copies as Appendix E. For a probability vector π on a supplied block K , define exact exterior penalties

$$\rho_{uv}(\pi) = \sum_{w \notin K} |\pi_w B_{uw} - \pi_u B_{vw}|, \quad (14)$$

$$H_\pi = L(B_K - \rho(\pi)).$$

The copy expansion and metric inequalities give

$$F\left(\sum_{r \in K} \pi_r T_r(Y)\right) - F(Y) \geq \sum_{u < v \in K} (B_{uv} - \rho_{uv})(1 - Y_{uv}) = \frac{1}{2} \text{tr}(H_\pi Y_{KK}).$$

Thus $H_\pi \geq 0$ ensures an optimal fused representative. A positive restriction to $\mathbf{1}^\perp$ forces every relaxed optimum to fuse K . This is a specified improving-map sufficient test within the established framework [30].

Uniform copying uses $\pi_r = 1/|K|$. The full-pair degree-anchor reference uses $\pi_r = d_r/V$, for which

$$\rho_{uv} = \frac{d_u d_v}{V} \sum_{w \notin K} \left| \frac{A_{uw}}{d_u} - \frac{A_{vw}}{d_v} \right| \leq \frac{d_u d_v}{V} (\delta_u + \delta_v) = h_{uv}.$$

Consequently $H_\pi = H + L(h - \rho)$, where $H = L(B_K - h)$ and the second Laplacian has nonnegative weights. The full-pair test contains the degree/profile sufficient test on the same block and

graph. This pointwise relation does not prescribe the final quotient produced by different sequences of weak choices.

For $K = \{u, v\}$, let p be the probability of copying to u . The optimized pair loss is

$$s(p) = \sum_{w \notin \{u, v\}} |p(B_{uw} + B_{vw}) - B_{uw}|, \quad 0 \leq p \leq 1.$$

The threshold $B_{uv} \geq \min_p s(p)$ guarantees weak fusion, and strict inequality guarantees fusion in every relaxed optimum. Nonzero terms have breakpoint $B_{uw}/(B_{uw} + B_{vw})$ and weight $|B_{uw} + B_{vw}|$; zero sums contribute constants. The implementation takes the lower weighted median of all breakpoints, then clips it to $[0, 1]$; it chooses $p = 0$ if every slope is zero. This optimizes constant mixtures of these two feasible copies, including uniform, degree and deterministic-anchor choices. It is not full discrete Rule5 or a general anchor/SDP-dual optimization procedure.

Every accepted weak operation is rechecked on its current quotient. In particular, endpoint mixtures may support only one direction, so arbitrary unions of weak pair facts are avoided. Membership congruence represents the fused face exactly: $Y = CZC^\top$, $Z \in \mathcal{M}_q$. The principal representative submatrix proves the reverse inclusion. With $B' = C^\top BC$, this gives $\langle B, Y \rangle / S = \langle B', Z \rangle / S$, including loop and internal diagonal mass. Sequential relaxed-safe stages therefore preserve the full optimum.

Positive signed-support components supply the common pre-solve. Every cross-component coefficient is nonpositive and every feasible co-membership is nonnegative. Replacing a feasible matrix by the direct sum of its principal component matrices cannot decrease the objective. That direct sum preserves PSD, diagonals, boxes and all triangles: a triple crossing components reduces to a box constraint or to zero. Hence the plain metric optimum is the sum of the component optima in original normalization. General anchor optimization, a general relaxed-map verifier and generic discrete cut/Rule4/Rule5 extensions are outside the implemented comparison.

F.2 Fixed inputs, numerical policy and exact proofs

The metric study uses the original 43 case dictionaries, five arms and fixed solver policies in its separate configuration. The 39 controlled cases include 35 confirmatory cells, three proved-failure stresses and one derived negative control; the other four graphs were previously used for development. It does not pool its measurements with the six feasibility pilots, discrete V1/native comparisons or public V2 Louvain workloads. The legacy uniform reference's 150-vertex exclusion does not apply here: the relaxation-safe uniform control operates on the fixed metric inputs, whose original sizes are at most 300. Each arm follows its own sequential weak operations on its actual current quotient and retains composed original memberships.

The solver models PSD, unit diagonals, entry bounds and active metric cuts. Each separation round adds at most 5,000 violated oriented triangles in descending violation order with lexicographic ties, using threshold 10^{-7} ; cuts are never removed, and at most 50 rounds are allowed. SCS starts with accuracy 10^{-6} and tightens to 10^{-7} and then 10^{-8} only when no new cut is found and

the exact interval remains too wide. Each solve has an iteration cap of 10^6 and normalization enabled. Only the canonical primal proposal warms the next solve; slacks and duals are reset. Pilot eligibility requires completing the 240-vertex sentinel. Among eligible backends, the fixed rule first maximizes completed sentinels, then minimizes summed compute cost with 180 seconds assigned to unresolved pilots, and finally favors direct on an exact tie. Both backends completed all three sentinels, and direct was selected by the second criterion.

For each signed-support component, exact coefficients retain original S and resolution. Primal proposals, free diagonal dual variables and inequality multipliers are quantized with denominator 2^{40} ; negative inequality multipliers are clipped to zero. A rational identity shift and rescaling enforce primal PSD, entry bounds and all triangles. A rational Gram factor plus a diagonally dominant residual supplies the dual correction, with zero multipliers for inactive cuts. Original multiplicities allocate component widths whose sum is at most $1/1000$. Singleton and zero-objective components use exact analytic bounds. Component records retain numerical proposals, quantization, factors and exact objective arithmetic. The general rigorous-bound idea is established [29]; these checks authenticate this implementation's particular proposals.

Original CSR, full degrees, candidate bank and incumbent are checkpointed before composition. Synchronous durable events follow every validated accepted operation, and their I/O belongs to preprocessing time. An interrupted trace records a valid prefix, without asserting a completed fixed point. Imports and input decoding have a separate initialization guard; final serialization has a separate bounded window. Both contribute to physical process wall and monitored RSS, and either failure prevents target credit. Time ratios never replace an unresolved job by its budget.

An independent post-measurement audit checks original inputs, current-quotient acceptance for every recorded operation, composed memberships and components, raw proposal/quantization bindings, exact primal and dual proofs, original lift feasibility and transferred bounds. Its inventory accounts for all 215 planned outcomes and the operator's command, output and exit records. The descriptive analysis accepts scientific target credit only after that complete audit; diagnostic solver statuses alone are insufficient.

G PROFILE-PENALTY ENVELOPE

This section quantifies the spectral overestimate introduced by the center. It specializes established L^1 cut representation [34] and product-demand Schur energy minimization [35, Definitions G.13–G.14 and the proof of Lemma C.2]. We give the algebra explicitly. The weighted-center result is a mathematical variant; every measured D arm retains the unweighted production median.

For positive full degrees on K , set $V = \sum_{i \in K} d_i$ and $a_i = (A_{iw}/d_i)_{w \notin K}$, including implicit zeros. For a fixed center c , let $\delta_i = \|a_i - c\|_1$ and define

$$\begin{aligned} \rho_{ij} &= \frac{d_i d_j}{V} \|a_i - a_j\|_1, \\ h_{ij} &= \frac{d_i d_j}{V} (\delta_i + \delta_j), \\ R_e &= L(\rho), \quad P_c = L(h). \end{aligned} \tag{15}$$

Thus $\rho_{ij} = p_{ij}^{\text{ext}}$ from Eq. (4). Both Laplacians have adjacency-weight units. The inequalities below hold on all real vectors, with the convention $x^\top L(w)x = \sum_{i < j} w_{ij}(x_i - x_j)^2$.

PROPOSITION 2 (MEDIAN ENVELOPE). *If c is a coordinatewise degree-weighted median, then*

$$R_e \leq P_c \leq 3R_e.$$

For the implemented unweighted median, with $k = |K|$ and $\kappa = \max_{i \in K} d_i / \min_{i \in K} d_i$, a valid bound is

$$R_e \leq P_c \leq C_{k,\kappa} R_e \leq (1 + 2\kappa) R_e, \\ C_{k,\kappa} = 1 + 2\kappa \frac{\lfloor k/2 \rfloor}{\lceil k/2 \rceil}. \quad (16)$$

PROOF. Triangle inequality gives $h_{ij} \geq \rho_{ij}$, proving the lower bound. For a nonempty proper side $T \subset K$, let $p = \sum_{i \in T} d_i / V$. Let C_T have weights $d_i d_j / V$ across $T \mid T^c$, and J_T have these weights only within T . For an arbitrary vector x , let μ_T, μ_{T^c} be its weighted side means and $E_T = \sum_{i \in T} d_i (x_i - \mu_T)^2$, with E_{T^c} defined similarly. Weighted-variance expansion gives

$$x^\top J_T x = p E_T, \\ x^\top C_T x = (1 - p) E_T + p E_{T^c} \\ + p(1 - p) V (\mu_T - \mu_{T^c})^2. \quad (17)$$

Consequently

$$J_T \leq \frac{p}{1 - p} C_T, \quad C_T + 2J_T \leq \frac{1 + p}{1 - p} C_T.$$

Equivalently, eliminating T^c in the product-demand biclique gives

$$\text{Sc}_{T^c}(C_T) = \frac{1 - p}{p} (J_T)_{TT},$$

and minimizing the full cut energy over x_{T^c} gives the same inequality. Here the Schur subscript denotes the eliminated indices. Both sides have positive mass; the eliminated block $p \text{diag}(d_{T^c})$ is invertible.

For each coordinate ℓ and $t \geq 0$, take $T_{\ell,t}^+ = \{i : a_{i\ell} > c_\ell + t\}$ and $T_{\ell,t}^- = \{i : a_{i\ell} < c_\ell - t\}$. Layer-cake decomposition expresses the scalar pair distance as

$$|a_{i\ell} - a_{j\ell}| = \int_0^\infty (|1_{T_{\ell,t}^+}(i) - 1_{T_{\ell,t}^+}(j)| \\ + |1_{T_{\ell,t}^-}(i) - 1_{T_{\ell,t}^-}(j)|) dt.$$

The center distance sum has the same integral with each absolute indicator difference replaced by its indicator sum. Hence its atom is $C_T + 2J_T$, whereas the pair-distance atom is C_T . An empty side contributes zero. A full side has zero cut but generally nonzero center energy and is excluded by the median premise.

For a weighted median, each strict threshold side has $p \leq 1/2$. For an unweighted median, it has at most $\lfloor k/2 \rfloor$ indices, so

$$\frac{p}{1 - p} \leq \kappa \frac{\lfloor k/2 \rfloor}{\lceil k/2 \rceil}.$$

Summing coordinates and integrating the nonnegative atom inequalities proves both upper bounds. This also covers midpoint medians for even k . $\square$

For an arbitrary center, define τ as the maximum degree-volume fraction strictly above or strictly below it in any coordinate, with $\tau = 0$ if there are no coordinates. If $\tau < 1$, the same proof gives $R_e \leq P_c \leq (1 + \tau)(1 - \tau)^{-1} R_e$. Here τ is a one-sided volume fraction. Identical profiles and a different center have $R_e = 0$ and nonzero P_c , so unrestricted centers admit no finite universal factor.

The universal weighted-median factor 3 is sharp. For one binary coordinate equal to one on T and zero elsewhere, choose center zero and a nonconstant x_T with weighted mean zero, setting $x_{T^c} = 0$. With at least two indices in T , Eq. (17) gives ratio $(1 + p)/(1 - p)$: $p = 1/2$ attains 3 with an endpoint median, while unique medians at $p < 1/2$ approach it. For the production midpoint policy, the odd family $k = 2r + 1$, $r \geq 2$, with r one-profiles of weight κd_0 and $r + 1$ zero-profiles of weight d_0 has unique median zero and ratio $1 + 2\kappa r/(r + 1)$. This approaches $1 + 2\kappa$ over unrestricted sizes; we make no optimal-factor claim for the cap64 policy. For $k = 2$, a center between the two coordinate values instead gives $h_{12} = \rho_{12}$ exactly. These are symbolic sharpness families, rather than measured graph cases.

Finally let $G = L(B_K)$. If $P_c \leq CR_e$, then

$$G - CR_e \leq G - P_c \leq G - R_e.$$

Thus $G - CR_e \geq 0$ suffices for center acceptance; positivity on $1^\perp$ suffices for strict acceptance. This is a robust full-pair premise, stronger than merely $G - R_e \geq 0$. The center condition is exactly Theorem 1's projected condition, by Appendix A.2. The envelope bounds exterior-penalty energy, and supplies no multiplicative guarantee for a signed margin, resolution threshold, accepted coverage or running time.

H NECESSARY STRUCTURE OF THE FIRST SD-D MERGE

Use the exact S and D controls in Appendix F. S scans every non-negative signed pair with optimized constant-mixture loss, then every eligible uniform collective bank block, restarting after each accepted merge. SD reaches a completed S fixed point before D starts. Both stages use the same original bank mapped and deduplicated before cap64, with exact current graph, memberships, full degrees, original S and resolution.

PROPOSITION 3 (FIRST ADDITIONAL D BLOCK). *If S completes its fixed point and D subsequently accepts a first block K on that unchanged quotient, then $|K| \geq 3$ and the positive current full degrees in K are not all equal. Weak acceptance suffices.*

PROOF. D excludes nonpositive block degrees and its bank excludes singletons. For an accepted pair $\{u, v\}$, its two-vertex Laplacian condition gives $B_{uv} \geq h_{uv} \geq \rho_{uv} \geq 0$. The optimized S pair loss is

$$s(p) = \sum_{w \notin \{u, v\}} |p B_{vw} - (1 - p) B_{uw}|.$$

At the included probability $p = d_u/(d_u + d_v)$, null cancellation gives

$$s(p) = \frac{d_u d_v}{d_u + d_v} \sum_{w \notin \{u, v\}} \left| \frac{A_{uw}}{d_u} - \frac{A_{vw}}{d_v} \right| = \rho_{uv}.$$

Thus $B_{uv} \geq \min_p s(p)$ and this nonnegative pair would pass S, including at equality, contradicting its fixed point.

If all block degrees equal $d > 0$, then $V = kd$ and

$$\begin{aligned}\rho_{ij} &= \frac{1}{k} \sum_{w \notin K} |A_{iw} - A_{jw}| \\ &= \frac{1}{k} \sum_{w \notin K} |B_{iw} - B_{jw}| = \rho_{ij}^{\text{uniform}}.\end{aligned}$$

Exterior degrees need not be equal. Since $h \geq \rho$, D acceptance implies

$$L(B_K - \rho^{\text{uniform}}) = L(B_K - h) + L(h - \rho^{\text{uniform}}) \geq 0.$$

Before D's first merge, the graph and membership map are unchanged. The same mapped/capped block would therefore pass S's uniform check, another contradiction. $\square$

The premise requires a genuinely completed S scan. An interrupted accepted prefix or phase field alone does not establish it. The statement applies to the first SD-D block, rather than standalone D or later D quotients, because SD never returns to S. It proves necessary joint structure if an additional acceptance occurs; existence, frequency and optimization utility require separate evidence. Stronger collective-anchor or general improving-map controls are outside this invariant.

H.1 A connected heterogeneous witness

The following exact construction separates the initial S and D tests on a connected positive signed support, even if S receives every original subset. It is an analytic witness outside the frozen 43-input metric study. Put $H = \{0, 1, 2\}$, $L = \{3, 4, 5\}$, $K = H \cup L$, and let W contain 20 further vertices. Set $\gamma = 375/2048$. The symmetric adjacency has

$$\begin{aligned}A_{ij} &= 295/256 & (i \in H, j \in L), \\ A_{iw} &= d_i/64 & (i \in K, w \in W), \\ A_{ii} &= 8267/256 & (i \in H), \\ A_{ii} &= 523/256 & (i \in L), \\ A_{ww} &= 211/16 & (w \in W),\end{aligned}$$

and every other off-diagonal entry is zero. Here $d_i = 52$ on H and $d_i = 8$ on L ; the displayed row sums give $d_w = 16$ on W , so $S = 500$, $V_K = 180$, and $\beta = \gamma/S = 3/8192$. Write $z = 5/512$ and, for distinct vertices of the indicated types,

$$\begin{aligned}B_{HL} &= 1, & B_{HH} &= -507/512 = -a_H, & B_{LL} &= -3/128 = -a_L, \\ B_{Hw} &= 65/128 = b_H = 52z, & B_{Lw} &= 5/64 = b_L = 8z, & B_{ww'} &= -3/32 = -b_W.\end{aligned}$$

Every positive signed edge is between types, and these edges connect all 26 vertices. Finally choose two distinct H vertices $0, 1$ and $w_1 \in W$. For $\varepsilon = 2^{-20}$ increase A_{0w_1} and A_{1w_1} by ε , decrease A_{00} and A_{11} by ε , and decrease $A_{w_1 w_1}$ by 2ε . Full degrees, S , γ , β and B_K are unchanged. All entries and the quotient after fusing K remain nonnegative, dyadic and exactly representable in binary64.

PROPOSITION 4 (SCOPED JOINT DEGREE WITNESS). *On the perturbed graph, every nonnegative signed pair fails S's optimized-pair test and every original subset of size at least two fails S's uniform test. With the supplied bank $\{K\}$, the first SD-D operation strictly accepts K after the completed S fixed point. Its normalized exterior profiles are unequal, and the actual unweighted-median penalty is*

strictly larger than the full-pair penalty on one internal pair. W nevertheless also accepts K . Before any contraction, the published discrete almost-clique Rule 4 fails on every nontrivial subset, and published similar-neighborhood Rule 5 fails on every positive pair. These are initial-criterion statements; they do not exclude arbitrary adaptive or exact-subproblem preprocessing.

PROOF. For H-L optimized pairs, the two other H columns each contribute at least a_H to the exterior loss for every mixture $p \in [0, 1]$, so the loss is at least $2a_H > 1 = B_{HL}$. For H-W, the two other H columns contribute at least b_H each, while its affinity is at most $b_H + \varepsilon < 2b_H$. For L-W, the 19 other W columns each contribute at least $\min(b_L, b_W) = b_L$, greater in total than $B_{Lw} = b_L$. All other pairs have negative affinity. Thus no optimized pair passes S.

For any core-only uniform block $J \subseteq K$ of size $r \leq 6$, the 20 exterior W columns give each H-L pair a penalty at least

$$\frac{20}{r}(b_H - b_L) \geq \frac{20}{6} \frac{55}{128} = \frac{275}{192} > 1 = B_{HL}.$$

The perturbation only enlarges these contrasts. Same-type internal weights are negative already, so every effective internal edge is negative. A W -only block also has negative internal weights and nonnegative penalties.

Now let J contain p vertices of H , ℓ of L , and $t \geq 1$ of W ; write $r = p + \ell \geq 1$, $c = b_W = 3/32$ and $C = z(52p + 8\ell)$. Before the perturbation, the singleton cut of a selected W vertex has energy no greater than

$$C - (t-1)c - \frac{20-t}{r+t}(C+rc). \quad (18)$$

We dropped other nonnegative exterior penalties. Multiplication by $r+t$ gives the numerator

$$N = C(r+2t-20) - c\{t(t-1) + 19r\}.$$

If $r+2t-20 \leq 0$, then $128N \leq -12\{t(t-1) + 19r\} \leq -228r \leq -228 < -285/4$. Otherwise replace C by its maximum for fixed r , obtained using up to three high-degree vertices. The following table gives the six resulting concave quadratics' discriminants after multiplication by 128:

r	$128C_{\max}(r)$	Discriminant
1	65	-50060
2	130	-60224
3	195	-30348
4	205	-23132
5	215	-14156
6	225	-3420

Every quadratic is negative for real t ; their largest maximum is the $r = 6$ value $-285/4$ after multiplication by 128. Thus the unperturbed cut is at most $-285/[512(r+t)] \leq -285/(512 \cdot 26)$. The two modified edges change its raw cut by at most 2ε and its averaged exterior penalty by at most 2ε . Since $4\varepsilon < 285/(512 \cdot 26)$, every mixed block still fails. These cases cover every original uniform subset, independently of the bank.

The actual normalized exterior row of each core vertex was $1/64$ on all 20 columns. After the perturbation, rows $0, 1$ take value $1/64 + \varepsilon/52$ at w_1 ; four core rows stay at $1/64$. Hence the ordinary unweighted coordinate median is still $1/64$, with $\delta_0 = \delta_1 = \varepsilon/52$,

all other deviations zero, and $\mu = \varepsilon/90$. In the notation of Appendix G, $h_{01} = 26\varepsilon/45$ whereas $\rho_{01} = 0$, since the two altered profiles are identical.

The unchanged signed internal Laplacian $L(B_K)$ has eigenvalues 15/512 on the two within- H zero-sum directions, 375/128 on the two within- L directions, and 6 on the group-difference direction. Moreover $\sum_{i<j} h_{ij} \leq \sum_i d_i \delta_i = 2\varepsilon$. As $(x_i - x_j)^2 \leq 2\|x\|_2^2$,

$$x^\top L(B_K - h)x \geq (15/512 - 4\varepsilon)\|x\|_2^2 > 0 \quad (x \perp \mathbf{1}).$$

Translation by a constant identifies this strict all-vector Laplacian condition with D's exact restricted condition on $d_K^\perp$, as in Appendix A.2. Thus D strictly accepts K . Since $\rho_{ij} \leq h_{ij}$, W accepts the same block. The prescribed bank $\{K\}$ maps to a singleton after the merge; this states a symbolic trajectory, not an observed run or a discovery-frequency claim.

For completeness, the published discrete almost-clique test [8] fails directly on supplied K : its positive internal graph is $K_{3,3}$ with unit edges and minimum cut 3, while its negative internal mass is $3a_H + 3a_L = 1557/512 > 3$. Its positive exterior boundary adds $1125/32 + 2\varepsilon$. In fact no original nontrivial subset passes Rule 4. If it contains no W , either its internal positive graph is disconnected or its cut is at most 3 while its positive W boundary is at least $20(b_H + b_L) > 3$. If it contains $1 \leq t \leq 19$ W vertices and some core vertex, a W singleton bounds its positive internal cut by $pb_H + \ell b_L$ (choose an unaltered W when $t \geq 2$); the positive boundary to omitted W is at least $(20-t)(pb_H + \ell b_L)$, and the negative internal W - W mass is $\binom{t}{2}b_W$. The $t = 1$ perturbation adds at most 2ε against nineteen boundary copies. A W -only block has zero positive internal cut and negative internal mass. At $t = 20$, an unaltered W singleton bounds the positive cut by $pb_H + \ell b_L \leq 225/128$, below its $190b_W = 285/16$ negative mass.

Rule 5 of the same published source maximizes over disjoint common-neighbor sets and includes the empty choice. On positive H- L edges its two exclusive-neighbor terms are $2+2a_H$ and $2+2a_L$; on H- W they are at least $19b_H + 2a_H$ and $2b_H + 19b_W$, while the edge weight is at most $b_H + \varepsilon$; on L- W the terms are $19b_L + 2a_L$ and $2b_L + 19b_W$. For each type their minimum already exceeds its signed edge weight, with strict slack surviving the perturbation. Thus the full published original-state pair test refuses, regardless of which common-neighbor subsets are later considered. $\square$

The instance has a connected positive signed support and nonzero observed profile dispersion. Its exact input weights are admissible to direct floating-CSR D verification with a supplied block. The current discovery front end rejects self-loops, so this looped witness is not an end-to-end input to that pipeline. The project's separate local Rule 4 checker requires integer adjacency; uniform scaling by 2^{20} preserves the signed inequalities and gives an integer encoding in principle, but that adapter and checker were not executed. The stronger full-pair W criterion contains D on the same graph and block, as shown by $h \geq \rho$. Published adaptive improving maps and incident-subinstance oracles remain unexcluded, and the synthetic loops and low resolution are limitations. This construction proves no running-time, discovery, or broad-method gain.

I ADDITIONAL DISCRETE PERFORMANCE AND ASSEMBLY RESULTS

I.1 Full public coverage

I.2 Separate first-round summary

The first-round standalone experiment gives the same qualitative overhead result, with all 54 warm aggregate medians below one and nine-call ratios 0.546–0.749. Its degree and recursive structural partitions are incomparable; positive median additional join rank occurs in 14 of 18 settings, reaching 1,376 identifications. That structural statistic does not itself justify a safe hybrid. Median checking cost is 0.144–2.275 seconds and complete preparation is 0.430–6.892 seconds. Appendix K retains these separate V1 observations, including full costs, cap exclusions and largest-component coverage. V2 executes and charges every pipeline stage; the two timing studies are not pooled.

I.3 Isolated assembly methods and diagnostics

Isolated processes measure median-center assembly, full-pair assembly and verification, and a dense floating exterior diagnostic on fixed-core graphs with 100,1,000,10,000 exterior vertices and block sizes 16,32. Three timing repetitions rotate method order. Separate memory repetitions enable allocation tracing; their timings are excluded. Peak process RSS includes graph input and preparation. Dense diagnostics never certify a contraction. This comparison isolates representation cost rather than declaring a dense assembly kernel to be a complete competitor. Figure 3 shows whole-checker time and traced allocation. At $k = 16$, median-certificate time increases from 1.48 to 1.65 ms as exterior size grows from 100 to 10,000; its traced peak stays near 0.063 MiB, while the dense diagnostic grows from 0.052 to 4.70 MiB. At $k = 32$, median-checker times are 6.02–6.31 ms, with traced peaks of 0.209–0.240 MiB; the dense diagnostic reaches 8.37 MiB. The smaller dense diagnostic can be faster because it performs no exact certification. These finite observations support the sparse representation claim, not asymptotic independence of the complete checker: the current CSR column slice uses an $O(n)$ index workspace.

J COMPLETE SEQUENTIAL-PIPELINE RESULTS

The following figures and table retain every setting of the separately frozen four-arm extension. Timing and exact modularity describe returned feasible solutions. The three discovery seeds are fixed observations, and the one-, three- and nine-call prefixes are correlated summaries. Observed ranges are not confidence intervals. Original-graph objective evaluation and explicit quality-matched denominators accompany all speedup comparisons.

K COMPLETE FIRST-ROUND TABLES

The following tables retain every declared public graph and resolution, and every controlled supplied-block family. Seed ranges are observed variation, not population confidence intervals. The complete CSV and JSON evidence includes raw statuses and all attempted comparisons.

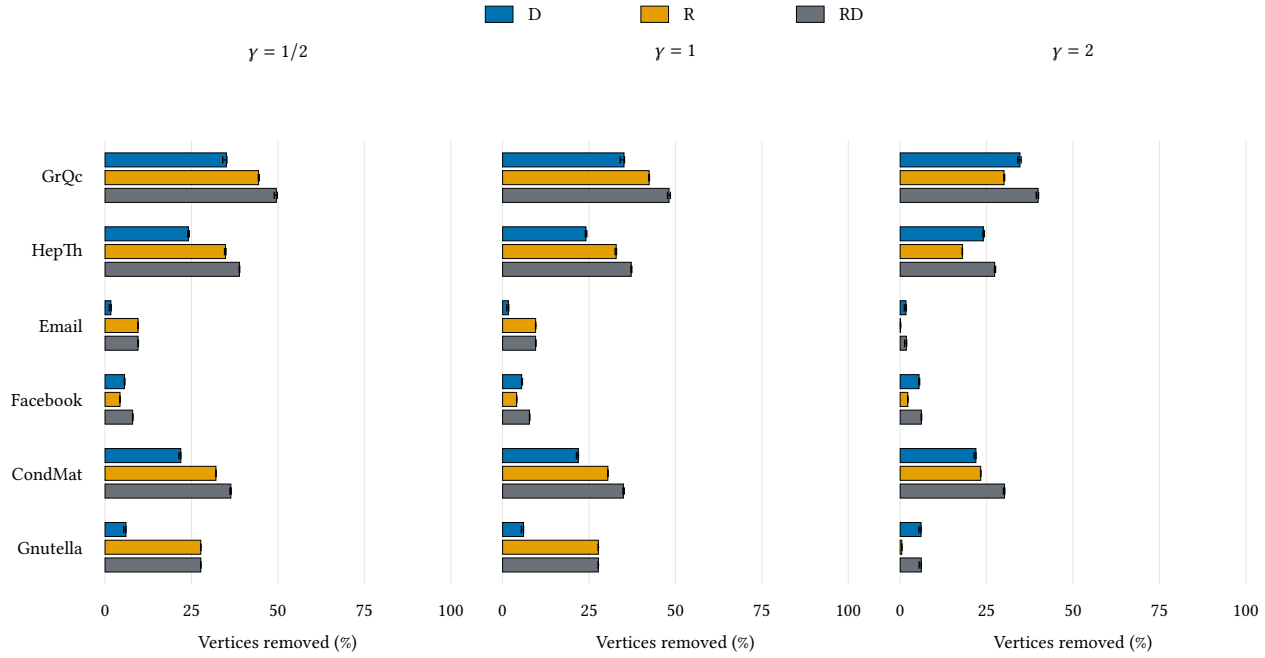


Figure 2: Complete V2 coverage on all six full graphs at all three resolutions. D, R and RD denote the specified public pipelines. Bars show median fractions of all original vertices removed over three fixed discovery seeds; whiskers are their observed minimum and maximum. All 18 graph/resolution groups are shown. RD executes its own fresh R prefix; its additional removals appear in Table 6 and Figure 4.

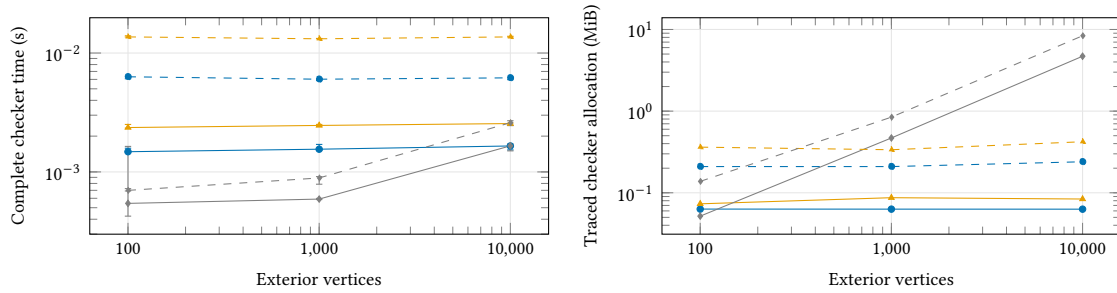


Figure 3: Fixed-core scaling. Circles: median certificate; triangles: full-pair certificate; diamonds: uncertified dense diagnostic. Time shows the median and range of three isolated, unprofiled repetitions; traced memory uses a separate repetition. Solid curves have $k = 16$ and dashed curves $k = 32$. Process RSS, including input and preparation, is reported separately.

Table 5: Complete V2 descriptive comparison summary. Each row contains all 18 graph/resolution settings and 54 fixed discovery-seed observations. Speedups are constructed standalone T_B/T_A ; the first range spans the 18 per-setting medians, and the count records medians above 1. Q counts are exact worse/equal/better outcomes for A versus B, out of 54. Matches require exact equal Q; their range uses every matched seed speedup, with its own non-null n and no values if $n = 0$. Prefixes k are correlated summaries of the same nine calls. These are observed counts, not pooled population inference.

Pair	k	Speedup range	> 1/18	Q worse/equal/better	Matches/54	Matched range (n)
D/U	1	0.19–0.33	0/18	32/7/15	7/54	0.18–0.28 (7)
D/U	3	0.32–0.54	0/18	31/4/19	4/54	0.33–0.41 (4)
D/U	9	0.55–0.75	0/18	31/3/20	3/54	0.59–0.62 (3)
R/U	1	0.13–0.33	0/18	21/6/27	6/54	0.20–0.24 (6)
R/U	3	0.21–0.55	0/18	29/4/21	4/54	0.28–0.39 (4)
R/U	9	0.38–0.75	0/18	25/2/27	2/54	0.40–0.63 (2)
RD/U	1	0.10–0.25	0/18	26/7/21	7/54	0.10–0.19 (7)
RD/U	3	0.17–0.43	0/18	34/4/16	4/54	0.23–0.32 (4)
RD/U	9	0.32–0.64	0/18	32/2/20	2/54	0.37–0.56 (2)
D/R	1	0.92–2.02	13/18	32/4/18	4/54	0.92–1.38 (4)
D/R	3	0.91–1.84	12/18	31/4/19	4/54	0.91–1.45 (4)
D/R	9	0.93–1.54	11/18	34/2/18	2/54	0.95–1.53 (2)
RD/R	1	0.72–0.80	0/18	25/13/16	13/54	0.71–0.81 (13)
RD/R	3	0.75–0.86	0/18	26/12/16	12/54	0.74–0.86 (12)
RD/R	9	0.81–0.92	0/18	26/11/17	11/54	0.80–0.93 (11)

Table 6: All 18 V2 graph/resolution coverage settings. Entries are median [minimum,maximum] over three fixed discovery seeds. The denominator is every original vertex. RD additional counts use its own freshly executed R prefix. The observed ranges are not confidence intervals.

Graph	γ	D removed (%)	R removed (%)	RD removed (%)	RD additional vertices
GrQc	1/2	35.1 [34.0,35.3]	44.4 [44.3,44.6]	49.5 [48.9,49.8]	272 [237,273]
GrQc	1	35.1 [34.0,35.3]	42.4 [42.3,42.5]	48.1 [47.7,48.5]	308 [278,317]
GrQc	2	34.7 [34.0,35.0]	30.0 [30.0,30.2]	39.9 [39.4,40.0]	520 [479,524]
HepTh	1/2	24.1 [23.9,24.4]	34.8 [34.6,35.0]	38.9 [38.8,39.0]	393 [392,426]
HepTh	1	24.1 [23.9,24.4]	32.9 [32.6,33.0]	37.2 [37.1,37.4]	437 [421,461]
HepTh	2	24.1 [23.9,24.4]	18.0 [18.0,18.0]	27.3 [27.3,27.6]	921 [921,944]
Email	1/2	1.7 [1.2,1.8]	9.6 [9.6,9.6]	9.6 [9.6,9.7]	0 [0,1]
Email	1	1.7 [1.2,1.8]	9.6 [9.6,9.6]	9.6 [9.6,9.7]	0 [0,1]
Email	2	1.7 [1.2,1.8]	0.1 [0.1,0.1]	1.8 [1.3,1.9]	17 [12,18]
Facebook	1/2	5.6 [5.6,5.7]	4.3 [4.3,4.3]	8.0 [8.0,8.1]	147 [147,151]
Facebook	1	5.5 [5.5,5.7]	4.1 [4.1,4.1]	7.8 [7.8,7.9]	150 [150,154]
Facebook	2	5.5 [5.3,5.6]	2.2 [2.2,2.2]	6.1 [6.1,6.2]	159 [157,162]
CondMat	1/2	21.9 [21.4,21.9]	32.1 [32.1,32.1]	36.4 [36.1,36.5]	987 [932,1022]
CondMat	1	21.9 [21.4,21.9]	30.5 [30.4,30.5]	35.0 [34.8,35.2]	1034 [1008,1102]
CondMat	2	21.9 [21.4,21.9]	23.3 [23.3,23.3]	30.2 [29.9,30.3]	1587 [1523,1618]
Gnutella	1/2	6.0 [5.4,6.1]	27.7 [27.7,27.7]	27.7 [27.7,27.7]	0 [0,0]
Gnutella	1	6.0 [5.4,6.1]	27.7 [27.7,27.7]	27.7 [27.7,27.7]	0 [0,0]
Gnutella	2	6.0 [5.4,6.1]	0.5 [0.5,0.5]	6.1 [5.5,6.1]	355 [316,356]

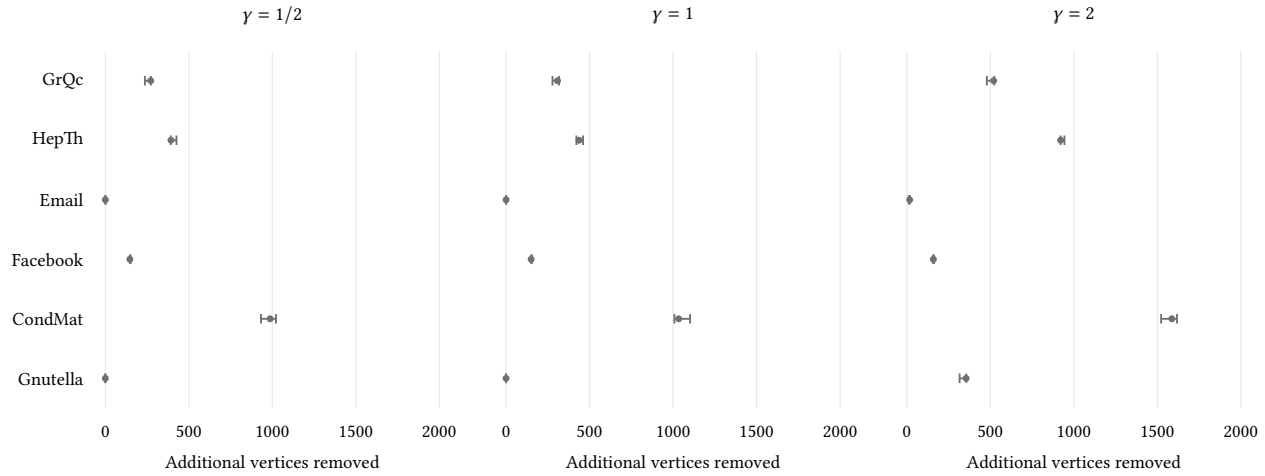


Figure 4: All RD additional vertex removals after its own freshly executed R prefix. Points show the median and whiskers the observed minimum and maximum over three fixed discovery seeds, on a shared vertex-count scale. These are not a structural-partition union or an extra solver run. Exact counts are retained in Table 6.

Table 7: Complete v1 public coverage. Entries are median [minimum,maximum] over three discovery seeds. D and R denote the degree certificate and the specified recursive criteria. Join ranks compare the structural partitions; they are not an executed safe hybrid pipeline.

Graph	γ	D removed (%)	R removed (%)	D beyond R	R beyond D
GrQc	1/2	35.1 [34.0,35.3]	44.4 [44.3,44.6]	216 [174,218]	715 [694,718]
GrQc	1	35.1 [34.0,35.3]	42.4 [42.3,42.5]	250 [209,257]	643 [618,647]
GrQc	2	34.7 [34.0,35.0]	30.0 [30.0,30.2]	463 [419,476]	220 [213,221]
HepTh	1/2	24.1 [23.9,24.4]	34.8 [34.6,35.0]	324 [322,347]	1397 [1356,1398]
HepTh	1	24.1 [23.9,24.4]	32.9 [32.6,33.0]	366 [353,380]	1232 [1191,1238]
HepTh	2	24.1 [23.9,24.4]	18.0 [18.0,18.0]	874 [868,893]	268 [264,284]
Email	1/2	1.7 [1.2,1.8]	9.6 [9.6,9.6]	0 [0,1]	79 [79,84]
Email	1	1.7 [1.2,1.8]	9.6 [9.6,9.6]	0 [0,1]	79 [79,84]
Email	2	1.7 [1.2,1.8]	0.1 [0.1,0.1]	17 [12,18]	1 [1,1]
Facebook	1/2	5.6 [5.6,5.7]	4.3 [4.3,4.3]	145 [143,149]	92 [88,96]
Facebook	1	5.5 [5.5,5.7]	4.1 [4.1,4.1]	147 [142,148]	83 [79,90]
Facebook	2	5.5 [5.3,5.6]	2.2 [2.2,2.2]	153 [149,159]	21 [20,22]
CondMat	1/2	21.9 [21.4,21.9]	32.1 [32.1,32.1]	733 [697,757]	3103 [3096,3175]
CondMat	1	21.9 [21.4,21.9]	30.5 [30.4,30.5]	796 [765,825]	2790 [2785,2865]
CondMat	2	21.9 [21.4,21.9]	23.3 [23.3,23.3]	1376 [1309,1393]	1703 [1701,1753]
Gnutella	1/2	6.0 [5.4,6.1]	27.7 [27.7,27.7]	0 [0,0]	1364 [1363,1404]
Gnutella	1	6.0 [5.4,6.1]	27.7 [27.7,27.7]	0 [0,0]	1364 [1363,1404]
Gnutella	2	6.0 [5.4,6.1]	0.5 [0.5,0.5]	355 [316,356]	4 [4,5]

		$\gamma = 1/2$			$\gamma = 1$			$\gamma = 2$		
D/U	GrQc	0.19	0.33	0.61	0.19	0.33	0.61	0.20	0.32	0.59
	HepTh	0.24	0.38	0.62	0.23	0.38	0.63	0.24	0.37	0.62
	Email	0.25	0.37	0.55	0.28	0.39	0.60	0.27	0.39	0.61
	Facebook	0.26	0.38	0.58	0.25	0.37	0.58	0.26	0.40	0.60
	CondMat	0.26	0.39	0.62	0.25	0.39	0.62	0.25	0.38	0.60
	Gnutella	0.26	0.38	0.58	0.27	0.41	0.62	0.33	0.54	0.75
		1	3	9	1	3	9	1	3	9
R/U	GrQc	0.21	0.36	0.65	0.20	0.37	0.63	0.21	0.34	0.61
	HepTh	0.22	0.37	0.63	0.24	0.39	0.67	0.23	0.37	0.61
	Email	0.19	0.29	0.48	0.20	0.30	0.48	0.20	0.32	0.53
	Facebook	0.13	0.21	0.38	0.13	0.21	0.38	0.13	0.22	0.39
	CondMat	0.22	0.36	0.61	0.22	0.35	0.61	0.22	0.35	0.57
	Gnutella	0.27	0.41	0.61	0.24	0.28	0.41	0.33	0.55	0.75
		1	3	9	1	3	9	1	3	9
RD/U	GrQc	0.16	0.30	0.57	0.16	0.29	0.56	0.16	0.27	0.52
	HepTh	0.18	0.30	0.55	0.18	0.31	0.57	0.17	0.28	0.51
	Email	0.14	0.22	0.39	0.15	0.23	0.40	0.15	0.23	0.42
	Facebook	0.10	0.17	0.32	0.10	0.17	0.33	0.10	0.17	0.33
	CondMat	0.18	0.29	0.54	0.17	0.28	0.54	0.17	0.28	0.50
	Gnutella	0.21	0.33	0.52	0.19	0.24	0.38	0.25	0.43	0.64
		1	3	9	1	3	9	1	3	9
D/R	GrQc	0.94	0.94	0.93	0.92	0.91	0.95	0.98	0.96	0.99
	HepTh	1.07	1.05	0.99	1.01	0.96	0.95	1.01	1.03	1.02
	Email	1.36	1.30	1.21	1.37	1.31	1.23	1.30	1.23	1.14
	Facebook	2.02	1.83	1.52	1.99	1.77	1.52	2.01	1.84	1.54
	CondMat	1.15	1.09	1.02	1.14	1.10	1.04	1.15	1.11	1.04
	Gnutella	0.93	0.93	0.96	1.17	1.46	1.53	0.97	0.98	0.99
		1	3	9	1	3	9	1	3	9
RD/R	GrQc	0.79	0.82	0.86	0.78	0.80	0.88	0.77	0.80	0.88
	HepTh	0.80	0.83	0.87	0.78	0.79	0.86	0.73	0.77	0.85
	Email	0.72	0.75	0.81	0.73	0.76	0.83	0.72	0.76	0.81
	Facebook	0.79	0.81	0.85	0.80	0.81	0.86	0.78	0.80	0.85
	CondMat	0.79	0.82	0.88	0.79	0.82	0.89	0.78	0.81	0.87
	Gnutella	0.76	0.80	0.86	0.80	0.86	0.92	0.76	0.78	0.85
		1	3	9	1	3	9	1	3	9

Correlated prefix k

Figure 5: All 270 V2 utility groups: five arm comparisons, all three resolutions, six full graphs and prefixes $k = 1, 3, 9$. Values are medians of constructed standalone T_B/T_A over three fixed discovery seeds. Blue exceeds 1 and orange is below 1; intensity uses one symmetric logarithmic scale across the full grid. A value rounded to unity retains a + or - direction marker when needed; n/a denotes an undefined median. Prefixes are correlated, not independent replications. No V1 or native timing is pooled.

		$\gamma = 1/2$			$\gamma = 1$			$\gamma = 2$		
D/U	GrQc	-2.9×10^{-4}	-7.0×10^{-4}	-8.5×10^{-4}	-4.2×10^{-4}	-6.0×10^{-4}	-4.8×10^{-4}	-2.4×10^{-3}	-1.6×10^{-3}	-1.0×10^{-3}
	HepTh	$+6.8 \times 10^{-4}$	$+8.6 \times 10^{-4}$	-1.8×10^{-4}	-1.5×10^{-3}	-1.4×10^{-3}	-7.2×10^{-4}	-4.3×10^{-4}	-1.9×10^{-3}	-1.2×10^{-3}
	Email	-2.1×10^{-3}	-2.7×10^{-4}	-1.2×10^{-4}	$+9.5 \times 10^{-4}$	$+8.0 \times 10^{-4}$	0	-7.0×10^{-4}	$+1.6 \times 10^{-4}$	$+1.6 \times 10^{-4}$
	Facebook	-1.7×10^{-5}	-8.5×10^{-5}	-7.0×10^{-5}	0	-8.6×10^{-5}	$+4.1 \times 10^{-4}$	-4.4×10^{-4}	-4.4×10^{-4}	-1.5×10^{-4}
	CondMat	$+6.8 \times 10^{-3}$	$+2.1 \times 10^{-4}$	-1.3×10^{-3}	0	$+5.2 \times 10^{-4}$	$+1.1 \times 10^{-3}$	-1.3×10^{-3}	$+2.5 \times 10^{-4}$	$+3.4 \times 10^{-4}$
	Gnutella	-7.1×10^{-3}	-4.3×10^{-3}	$+1.0 \times 10^{-3}$	0	0	0	$+5.3 \times 10^{-3}$	$+2.1 \times 10^{-3}$	$+8.1 \times 10^{-4}$
		1	3	9	1	3	9	1	3	9
R/U	GrQc	$+7.1 \times 10^{-4}$	$+2.5 \times 10^{-5}$	-7.5×10^{-5}	0	-6.1×10^{-4}	0	-1.9×10^{-3}	-1.3×10^{-3}	-1.2×10^{-3}
	HepTh	$+2.2 \times 10^{-4}$	$+1.9 \times 10^{-3}$	$+1.5 \times 10^{-4}$	-1.9×10^{-3}	-4.4×10^{-4}	$+3.3 \times 10^{-5}$	$+2.0 \times 10^{-4}$	-6.5×10^{-4}	-3.2×10^{-4}
	Email	-7.7×10^{-4}	$+6.9 \times 10^{-4}$	$+6.9 \times 10^{-4}$	0	-1.2×10^{-3}	$+2.8 \times 10^{-4}$	$+5.6 \times 10^{-5}$	-4.1×10^{-5}	$+9.2 \times 10^{-5}$
	Facebook	$+6.9 \times 10^{-4}$	-3.5×10^{-5}	-3.5×10^{-5}	$+1.9 \times 10^{-4}$	$+8.9 \times 10^{-6}$	$+8.9 \times 10^{-6}$	$+6.5 \times 10^{-5}$	$+1.2 \times 10^{-4}$	-7.9×10^{-5}
	CondMat	$+3.8 \times 10^{-3}$	$+1.4 \times 10^{-3}$	$+3.8 \times 10^{-4}$	0	0	$+1.3 \times 10^{-3}$	-1.1×10^{-3}	-4.0×10^{-4}	$+4.5 \times 10^{-4}$
	Gnutella	-1.5×10^{-2}	-1.5×10^{-2}	-1.6×10^{-2}	-4.1×10^{-4}	-4.1×10^{-4}	-5.2×10^{-4}	$+2.6 \times 10^{-3}$	$+1.9 \times 10^{-3}$	$+1.0 \times 10^{-3}$
		1	3	9	1	3	9	1	3	9
RD/U	GrQc	$+8.7 \times 10^{-4}$	-1.7×10^{-4}	-2.5×10^{-4}	0	-3.1×10^{-4}	0	-2.4×10^{-3}	-1.7×10^{-3}	-1.3×10^{-3}
	HepTh	$+9.0 \times 10^{-4}$	$+2.6 \times 10^{-3}$	$+9.7 \times 10^{-4}$	-1.5×10^{-3}	-4.9×10^{-4}	-1.5×10^{-4}	-4.1×10^{-4}	-1.2×10^{-3}	-1.0×10^{-3}
	Email	-7.7×10^{-4}	$+6.9 \times 10^{-4}$	$+6.9 \times 10^{-4}$	0	0	$+1.5 \times 10^{-4}$	$+6.3 \times 10^{-4}$	$+6.2 \times 10^{-4}$	$+2.3 \times 10^{-4}$
	Facebook	$+8.1 \times 10^{-4}$	-1.4×10^{-4}	-6.4×10^{-5}	$+1.0 \times 10^{-4}$	-8.1×10^{-5}	-8.1×10^{-5}	-9.2×10^{-4}	-2.6×10^{-4}	-2.8×10^{-4}
	CondMat	$+5.1 \times 10^{-3}$	$+1.5 \times 10^{-4}$	-8.6×10^{-4}	0	0	$+6.2 \times 10^{-4}$	-2.1×10^{-3}	-2.0×10^{-4}	$+4.0 \times 10^{-4}$
	Gnutella	-1.5×10^{-2}	-1.5×10^{-2}	-1.6×10^{-2}	-4.1×10^{-4}	-4.1×10^{-4}	-5.2×10^{-4}	$+1.5 \times 10^{-3}$	$+8.5 \times 10^{-4}$	$+1.3 \times 10^{-3}$
		1	3	9	1	3	9	1	3	9
D/R	GrQc	-1.0×10^{-3}	-7.2×10^{-4}	-7.8×10^{-4}	-7.8×10^{-4}	0	-1.5×10^{-4}	-4.4×10^{-4}	-2.2×10^{-4}	$+1.1 \times 10^{-4}$
	HepTh	$+4.6 \times 10^{-4}$	-1.1×10^{-3}	-1.1×10^{-3}	-5.8×10^{-5}	-1.0×10^{-3}	-1.2×10^{-3}	-6.0×10^{-4}	-1.3×10^{-3}	-1.0×10^{-3}
	Email	-1.3×10^{-3}	-9.6×10^{-4}	-8.1×10^{-4}	$+9.5 \times 10^{-4}$	$+2.0 \times 10^{-3}$	-8.1×10^{-5}	-7.6×10^{-4}	$+2.0 \times 10^{-4}$	$+6.4 \times 10^{-5}$
	Facebook	-7.0×10^{-4}	-5.1×10^{-5}	-3.5×10^{-5}	-1.9×10^{-4}	-9.5×10^{-5}	$+4.0 \times 10^{-4}$	-5.0×10^{-4}	-5.6×10^{-4}	-7.1×10^{-5}
	CondMat	-3.9×10^{-4}	-1.3×10^{-3}	-1.7×10^{-3}	0	0	-6.8×10^{-4}	-1.8×10^{-4}	$+5.9 \times 10^{-4}$	-1.1×10^{-4}
	Gnutella	$+8.0 \times 10^{-3}$	$+1.1 \times 10^{-2}$	$+1.7 \times 10^{-2}$	$+4.1 \times 10^{-4}$	$+4.1 \times 10^{-4}$	$+1.5 \times 10^{-3}$	$+2.7 \times 10^{-3}$	$+2.5 \times 10^{-4}$	-2.0×10^{-4}
		1	3	9	1	3	9	1	3	9
RD/R	GrQc	$+1.5 \times 10^{-4}$	-8.2×10^{-5}	-1.6×10^{-4}	-2.8×10^{-4}	$+2.4 \times 10^{-4}$	0	-4.8×10^{-4}	$+1.6 \times 10^{-4}$	-1.1×10^{-4}
	HepTh	$+2.7 \times 10^{-3}$	$+4.5 \times 10^{-4}$	$+8.1 \times 10^{-4}$	-5.8×10^{-5}	-6.4×10^{-5}	-6.2×10^{-4}	-4.7×10^{-4}	-1.1×10^{-3}	-7.1×10^{-4}
	Email	0	0	0	0	0	0	$+5.7 \times 10^{-4}$	$+6.6 \times 10^{-4}$	$+1.3 \times 10^{-4}$
	Facebook	$+1.2 \times 10^{-4}$	-1.0×10^{-4}	-3.0×10^{-5}	-9.0×10^{-5}	-9.0×10^{-5}	-9.0×10^{-5}	-9.8×10^{-4}	-3.8×10^{-4}	-2.0×10^{-4}
	CondMat	$+1.3 \times 10^{-3}$	-1.6×10^{-3}	-1.2×10^{-3}	0	0	$+5.2 \times 10^{-5}$	-1.1×10^{-3}	$+8.8 \times 10^{-4}$	$+1.5 \times 10^{-4}$
	Gnutella	0	0	0	0	0	0	-1.1×10^{-3}	-1.0×10^{-3}	$+3.3 \times 10^{-4}$
		1	3	9	1	3	9	1	3	9
Correlated prefix k										

Figure 6: All 270 exact-Q comparison groups. Cells annotate the median returned $Q_A - Q_B$ over three fixed discovery seeds, rounded to two significant digits in scientific notation while preserving every nonzero sign. Only an exact zero is printed as 0. Blue, white and orange encode positive, exactly zero and negative medians respectively, without encoding magnitude. All exact rational seed values and ranges remain in the canonical source CSV. Prefixes summarize the same nine downstream calls and are correlated.

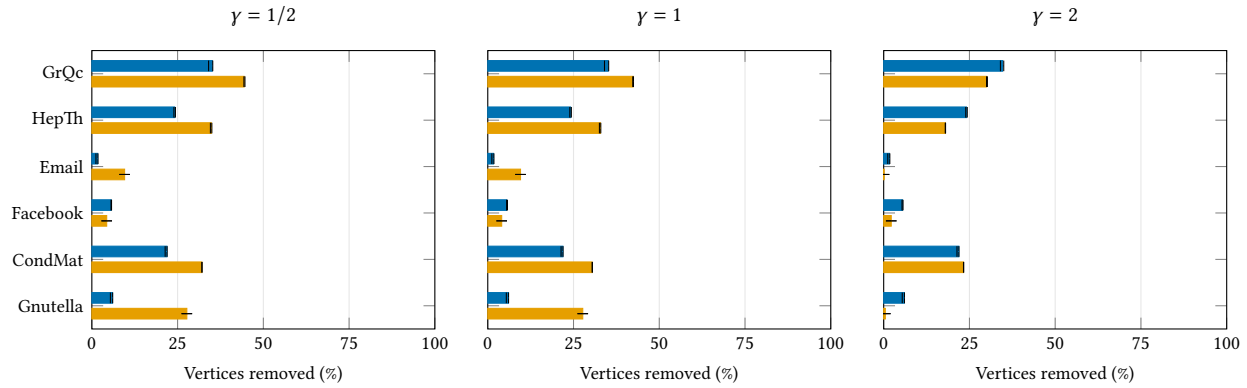


Figure 7: Public coverage. Blue: degree-weighted certificate; orange: recursive named criteria on the same bank. Bars show median removed fractions over three discovery seeds; whiskers show the observed range. All vertices remain in the denominator.

	$\gamma = 1/2$			$\gamma = 1$			$\gamma = 2$		
	1	3	9	1	3	9	1	3	9
GrQc	0.17	0.31	0.57	0.17	0.31	0.57	0.17	0.30	0.56
HepTh	0.20	0.35	0.59	0.20	0.36	0.60	0.21	0.35	0.60
Email	0.21	0.35	0.55	0.23	0.36	0.57	0.21	0.34	0.56
Facebook	0.21	0.33	0.56	0.22	0.34	0.55	0.22	0.37	0.55
CondMat	0.22	0.37	0.60	0.23	0.38	0.61	0.22	0.36	0.60
Gnutella	0.23	0.36	0.57	0.26	0.39	0.61	0.30	0.52	0.75

Figure 8: Cost recovery: median matched warm unreduced-to-reduced total-time ratio over three discovery seeds. Values above 1 recover the full reduction cost; values below 1 retain the overhead. Blue denotes recovery and orange denotes overhead; color intensity uses a common symmetric logarithmic scale about 1.

Table 8: Complete v1 checking and preprocessing cost, cap exclusions, and largest-component coverage, as median [minimum,maximum] over three discovery seeds. Component coverage retains the original full-graph objective and volume. Pre-processing includes the declared original representation, discovery, refinement, checker, quotient and solver-interface work.

Graph	γ	Checker (s)	Preprocessing (s)	Cap exclusions	LCC removed (%)
GrQc	1/2	0.520 [0.507,0.538]	1.199 [1.187,1.213]	68 [64,70]	26.7 [25.3,26.9]
GrQc	1	0.518 [0.502,0.557]	1.199 [1.194,1.245]	68 [64,70]	26.7 [25.3,26.9]
GrQc	2	0.513 [0.509,0.516]	1.199 [1.193,1.209]	68 [64,70]	26.1 [25.3,26.6]
HepTh	1/2	0.899 [0.874,0.957]	2.488 [2.479,2.610]	174 [171,176]	18.2 [18.0,18.5]
HepTh	1	0.956 [0.910,0.988]	2.553 [2.505,2.634]	174 [171,176]	18.2 [18.0,18.5]
HepTh	2	0.875 [0.874,0.954]	2.528 [2.489,2.575]	174 [171,176]	18.2 [18.0,18.5]
Email	1/2	0.148 [0.127,0.157]	0.430 [0.407,0.449]	16 [14,18]	1.7 [1.2,1.8]
Email	1	0.144 [0.128,0.145]	0.437 [0.410,0.444]	16 [14,18]	1.7 [1.2,1.8]
Email	2	0.153 [0.129,0.204]	0.437 [0.410,0.504]	16 [14,18]	1.7 [1.2,1.8]
Facebook	1/2	0.648 [0.640,0.707]	2.044 [2.031,2.280]	85 [84,86]	5.6 [5.6,5.7]
Facebook	1	0.695 [0.640,0.728]	2.147 [2.075,2.214]	85 [84,86]	5.5 [5.5,5.7]
Facebook	2	0.689 [0.644,0.721]	2.140 [2.079,2.261]	85 [84,86]	5.5 [5.3,5.6]
CondMat	1/2	2.266 [2.242,2.447]	6.886 [6.774,7.065]	494 [485,509]	18.1 [17.5,18.1]
CondMat	1	2.248 [2.241,2.269]	6.855 [6.744,6.928]	494 [485,509]	18.1 [17.5,18.1]
CondMat	2	2.275 [2.259,2.293]	6.892 [6.848,6.923]	494 [485,509]	18.1 [17.5,18.1]
Gnutella	1/2	0.510 [0.508,0.530]	1.866 [1.832,1.877]	155 [144,158]	6.0 [5.4,6.0]
Gnutella	1	0.516 [0.499,0.519]	1.872 [1.828,1.877]	155 [144,158]	6.0 [5.4,6.0]
Gnutella	2	0.528 [0.507,0.575]	1.869 [1.866,1.903]	155 [144,158]	6.0 [5.4,6.0]

Table 9: Every controlled supplied-block family and resolution. Counts are certified blocks over denominator N; U denotes unavailable comparisons, not mathematical rejection. AUTO is the production median checker; Exact is its forced exact counterpart. Zero and Full pair are independent rational reference conditions. The uniform reference retains its frozen graph-size guard. Private leaves are exploratory stress controls.

Family	γ	N	AUTO	Exact	Zero	Full pair	Uniform
Matched cliques	1/2	14	14	14	14	14	14
Matched cliques	1	14	10	12	12	12	12
Matched cliques	2	14	0	0	0	0	0
Common hubs	1/2	2	2	2	2	2	2
Common hubs	1	2	2	2	1	2	2
Common hubs	2	2	0	0	0	0	0
Weighted blocks	1/2	9	9	9	9	9	7
Weighted blocks	1	9	9	9	9	9	3
Weighted blocks	2	9	6	6	6	6	0
Noisy blocks	1/2	192	110	110	110	116	0 (192 U)
Noisy blocks	1	192	88	88	88	90	0 (192 U)
Noisy blocks	2	192	50	50	50	51	0 (192 U)
Exterior profiles	1/2	6	6	6	6	6	2 (4 U)
Exterior profiles	1	6	6	6	4	6	2 (4 U)
Exterior profiles	2	6	4	4	3	4	0 (4 U)
Uniform advantage	1/8	1	0	0	0	0	1
Private leaves	1	3	0	0	0	0	0